

Trade Effects on Substance Abuse: Evidence from Colombia's Liberalization

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Abstract

This paper examines the impact of trade liberalization on substance abuse. I study Colombia's 2010 unilateral tariff reform (Decree 4114), with immediate cuts concentrated in manufacturing. Using a Bartik-style measure of local tariff exposure and a continuous-treatment difference-in-differences design, I link the reform to municipality-level rates of substance-abuse hospitalizations and emergency-room visits over 2009–2014 from administrative health records. Municipalities more exposed to the tariff cut experienced larger increases in substance abuse. After correcting for multiple testing across the nine ICD-10 categories, multiple-drug use is the only category whose response remains individually robust, with a weaker effect on sedatives; the aggregate effect is present for adults aged 25–44 and is larger and more precisely estimated for males than for females. The evidence is consistent with a labor-market mechanism: the reform contracted formal employment in exposed tradable sectors, and the substance-abuse increase concentrates in the contributory health-insurance regime that covers formal employees. Drug-supply expansion and selective migration are not supported by the data as alternative mechanisms. The findings document downstream health costs of trade reform where social-protection infrastructure is limited.

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1 Introduction

Substance abuse is a major global health issue. Alcohol- and drug-use disorders are among the leading contributors to years lived with disability and premature mortality worldwide (Degenhardt et al., 2018; Rehm et al., 2009). According to the United Nations Office on Drugs and Crime, about 1 in 20 adults (ages 15–64) used at least one drug in 2014, and nearly 12% of these individuals suffered from drug-related disorders, with overdose accounting for roughly one-third of drug-related deaths (UNODC, 2016). Although the burden of drug disorders is higher in more developed countries, little is known about how low- and middle-income countries cope with substance abuse. One potential pathway is labor disruption resulting from trade liberalization: in both developed and developing economies, trade shocks have been associated with job losses, lower wages, and transitions to less secure employment, particularly among less-skilled workers (Autor et al., 2013; Dix-Carneiro and Kovak, 2017).

This paper examines whether trade liberalization affects substance abuse among the working-age population in Colombia, a country with documented growth in drug consumption across the two decades preceding the reform window (Camacho González et al., 2016). Colombia is also a setting in which drug-related dynamics have historically been dominated by production, trafficking, and coca eradication on the supply side (Prem et al., 2023).¹ Using a continuous-treatment difference-in-differences design that exploits the 2010 unilateral tariff reform (Decree 4114), I find that municipalities more exposed to the cut experienced larger increases in substance-abuse hospitalizations and emergency-room visits relative to less-exposed areas. The reform’s immediate cuts were concentrated in manufacturing; cuts in mining tariffs followed mostly through the contemporaneous free trade agreements. The average effectively applied tariff fell from about 12% in 2010 to 8.5% in 2011 and 6.3% by 2014 (Bonilla-Mejía et al., 2026).²

The analysis uses administrative records from Colombia’s Ministry of Health, which capture hospitalizations and emergency-room visits with substance-related primary diagnoses (ICD-10 F10–F19) at all public and private health providers.³ The records

¹Coca is the plant from which cocaine is derived. Eradication refers to government-led aerial-spraying or manual-destruction campaigns aimed at reducing the area under coca cultivation.

²The effectively applied tariff is the lowest duty importers actually face on a given product. Section 5 gives the full definition.

³Chapter V of ICD-10 codes F10–F19 covers mental and behavioural disorders due to psychoactive substance use, ranging from alcohol (F10) to use of multiple substances (F19).

distinguish between unique patients (extensive margin of treatment) and total services delivered (overall treatment volume), and the universal mandatory-reporting requirement reduces concerns about self-selection into the data based on socioeconomic confounders.

The relationship between trade exposure and substance abuse plausibly operates through stress associated with employment and income shocks. Job loss has been associated with higher hospitalizations for alcohol-related conditions, self-harm, and traffic accidents (Browning and Heinesen, 2012; Keefe et al., 2002); with increased mortality from suicide and substance-related causes (Eliason and Storrie, 2009); and with effects that intensify under prolonged unemployment (Classen and Dunn, 2012). Even the risk of layoffs has been linked to worse mental health among workers and their households, and to greater use of antidepressants and cardiovascular medications (Bubonya et al., 2017; Bünnings et al., 2017; Jolly, 2020; Colantone et al., 2019; Hummels et al., 2016). The link from labor-market stress to substance abuse specifically, as distinct from broader stress-related outcomes, has been documented most prominently in the “deaths of despair” literature for the United States, which traces rising mortality from alcohol, suicide, and drug overdoses to economic decline in affected communities (Case and Deaton, 2020).⁴

I find that, after the 2010 reform, municipalities more exposed to the tariff cut experienced larger increases in substance-abuse rates. Evaluated at the population-weighted average reduction in the local tariff exposure measure between 2010 and 2014 of 3.56 percentage points, the baseline estimates imply increases of roughly 44% in the patient rate and 56% in the service rate relative to the pre-reform mean. The dynamic event-study specification is consistent with parallel pre-reform trends and shows that the response builds up gradually over two to three years. After accounting for multiple testing across the nine ICD-10 substance categories with sharpened q -values,⁵ multiple-drug use is the only category whose response remains individually robust, with a weaker response for sedatives; the remaining categories are negatively

⁴Another body of research documents that mortality and several risky behaviors, including alcohol consumption and dangerous driving, were procyclical, rising during expansions when the opportunity cost of leisure is high (Ruhm, 2000). Ruhm (2015) shows that this procyclical pattern has weakened in recent decades, with several causes of death now showing limited or countercyclical responses to economic fluctuations.

⁵Sharpened q -values are an Anderson-style adjustment that controls the false-discovery rate across a family of related outcome regressions (Anderson, 2008).

signed but do not survive the correction. The aggregate response is therefore broader than the surviving categories alone: rescaling the multiple-drug coefficients implies they account for under a third of the total effect, so the magnitude is not concentrated in the surviving categories. The response is statistically significant for adults aged 25–34 and 35–44 and indistinguishable from zero for the youngest (18–24) and oldest (45–65) groups, and is larger and more precisely estimated for males than for females, consistent with the higher representation of men in the affected tradable sectors.

The results are robust to alternative control sets, sample restrictions, outcome transformations, and Bartik-sensitivity diagnostics. A placebo on acute-condition diagnoses that share the same reporting channel rules out a reporting-quality explanation, and a most-favored-nation reconstruction of the treatment isolates the response to the 2010 unilateral cut rather than the contemporaneous free trade agreements.

I trace the substance-abuse response to a labor-market mechanism. Using household-survey data at the department–semester level, I find that the 2010 reform contracted formal employment in the most exposed tradable sectors, with no offsetting expansion in informal employment within the same sector and no meaningful wage adjustment in the short run. The dynamic profile of the formal-employment response parallels the gradual buildup observed for substance abuse. Splitting the substance-abuse outcome by health-insurance regime, I find that the response is concentrated in the contributory regime, which enrolls formal employees and self-employed contributors, and is statistically indistinguishable from zero in the subsidized regime, which covers people in conditions of vulnerability outside formal employment. The substance-abuse response therefore concentrates in the population segment most exposed to the formal-employment shock, although the estimates remain local exposure effects on the working-age population in each municipality rather than individual treatment effects on workers in liberalized industries. Two alternative mechanisms are rejected by the data. First, tariff exposure does not predict coca cultivation or eradication, and the substance-abuse response is essentially unchanged when historically coca-cultivating municipalities are excluded, ruling out the supply-side channel highlighted by Colombia’s drug-market context. Second, the working-age population does not shift across municipalities with tariff exposure, ruling out selective migration.

These magnitudes are economically meaningful. A clean head-to-head compari-

son with the developed-country trade-and-health literature is not possible because the cited studies measure different objects in different units: [Pierce and Schott \(2020\)](#) report drug-mortality elasticities to NTR-gap exposure in U.S. counties after the granting of permanent normal trade relations to China,⁶ while [Adda and Fawaz \(2020\)](#) report hospitalization and mortality responses to import competition in U.S. commuting zones. The Colombian estimate here is a hospitalization-and-ER response per percentage-point tariff cut at the municipality level. The shared qualitative result is that locally concentrated trade shocks raise the demand for acute-condition health services; the absolute magnitudes are not directly comparable across the three settings. The estimates likely represent a lower bound on the true effect because the administrative records capture only hospitalizations and emergency-room visits and miss the population that never seeks treatment.

This paper speaks to broader debates on the social consequences of trade liberalization in low- and middle-income countries. The literature has documented effects on productivity, prices, wages, and sectoral reallocation ([Eslava et al., 2004](#); [Goldberg and Pavcnik, 2005](#); [Kovak, 2013](#)); less attention has been paid to the downstream effects on public health and social vulnerability. Understanding these effects is central to development policy in settings where labor-market adjustment programs, addiction-treatment systems, and social protection are limited.

This paper makes three contributions to the literature reviewed in Section 2. First, to my knowledge it is the first paper to link trade policy to substance abuse in a country whose drug-related dynamics are dominated by supply-side activity, using administrative health records coded by international diagnostic standards and disaggregated by substance category, sex, and age group. Second, it traces the substance-abuse response to a labor-market mechanism, using the cross-sectoral dispersion of formal-employment effects and the regime-split pattern in the substance-abuse data to align the response with the population segment most exposed to the formal-employment shock. Third, it extends the trade-and-health literature to a substance-abuse outcome and to a setting where the active drug-supply economy could plausibly have masked or reversed the consumption-side response.

⁶Permanent normal trade relations were granted by the United States to China in 2000, eliminating the annual congressional review of China's most-favored-nation status and locking in low tariff rates on Chinese imports.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes Colombia’s trade liberalization, the 2010 tariff reform, and the healthcare system that produces the records. Section 4 presents the empirical strategy. Section 5 describes the data sources. Section 6 presents the empirical results, robustness checks, and mechanism analysis. Section 7 concludes.

2 Related literature

This section positions the paper against two literatures: the labor-market consequences of trade liberalization in developing countries, with attention to Colombia, and the trade-and-health link, with emphasis on substance abuse and developing-country settings.

2.1 Trade liberalization and labor markets

A body of work has documented the labor-market effects of trade liberalization in Colombia and other developing countries. [Goldberg and Pavcnik \(2005\)](#) show that tariff reductions during Colombia’s 1980s and 1990s reforms reduced industry wage premiums, with disproportionate effects on less-skilled workers. [Goldberg and Pavcnik \(2003\)](#) find that the same reforms increased informality in sectors facing larger tariff cuts, and [Goldberg and Pavcnik \(2007\)](#) document increases in urban poverty in the short run through wage and employment channels. [Eslava et al. \(2004\)](#) examine the productivity effects of Colombia’s structural reforms, finding evidence of reallocation toward more productive firms alongside displacement of workers.

For the more recent period covered in this paper, [Mejía et al. \(2018\)](#) study labor fluidity around the 2010 unilateral cut,⁷ and [Bonilla-Mejía et al. \(2026\)](#) document heterogeneous wage-bill responses across sectors around the Colombia–United States free trade agreement, with gains concentrated in services and weaker or negative effects in manufacturing and agriculture.

Outside Colombia, [Kovak \(2013\)](#) uses regional variation in trade exposure, an approach I adopt in this paper, to study Brazil’s trade liberalization and documents

⁷Labor fluidity denotes the rate at which workers and jobs move across employers or industries, typically measured by gross worker or job-flow rates.

persistent wage declines in more exposed regions. [Dix-Carneiro \(2014\)](#) shows that labor-market adjustment to trade shocks in Brazil was slow, with transition costs concentrated among less-skilled workers. [Dix-Carneiro and Kovak \(2023\)](#) survey the broader evidence on globalization and inequality across Latin America. While this literature establishes the labor-market consequences of trade reforms, the downstream effects on health, and on substance abuse in particular, remain largely unexplored.

2.2 Trade, health, and substance abuse

A separate literature has examined the health consequences of trade-induced labor-market disruptions, primarily in developed countries. [Case and Deaton \(2020\)](#) trace rising U.S. mortality from alcohol, suicide, and drug overdoses to economic decline in affected communities. [Pierce and Schott \(2020\)](#) find that U.S. counties more exposed to the granting of permanent normal trade relations to China experienced increases in drug-related mortality, and [Adda and Fawaz \(2020\)](#) document that import competition raised hospitalization and mortality rates in affected commuting zones. Related work links trade exposure or labor-market downturns to mental distress ([Colantone et al., 2019](#)), broader employment-health outcomes ([Lang et al., 2019](#); [McManus and Schaur, 2016](#)), and opioid abuse ([Carpenter et al., 2017](#)).

The evidence from developing-country settings is thinner and more heterogeneous. [Feng et al. \(2021\)](#) document health improvements following export-tariff reductions in China, operating through income and nutritional channels. [Panda \(2020\)](#) finds that trade liberalization reduced infant mortality across African countries. [Fernández Guerico \(2021\)](#) shows that trade-induced job loss in Mexico raised mortality from diabetes and lowered mortality from ischemic heart disease and chronic pulmonary disease as alcohol and tobacco use fell with incomes, illustrating that the trade-induced health response in a developing country can run in either direction depending on the outcome and the channel.

Beyond trade-specific shocks, a parallel literature in epidemiology and public health examines how macroeconomic crises affect population health and substance use. [Stuckler et al. \(2009\)](#) document that economic downturns in Europe were associated with rising suicide and alcohol-related mortality. [Tapia Granados and Diez Roux \(2009\)](#) find that mortality in the United States moved procyclically during the Great Depres-

sion, with suicide as the main exception. [Cutler et al. \(2002\)](#) extend this evidence to Latin America, showing that the Mexican financial crises of the 1980s and 1990s raised mortality among the very young and the elderly. Within Latin American substance-use epidemiology, [Medina-Mora et al. \(2023\)](#) document the burden of mental and substance-use disorders in Mexico over 1990–2021, and [Posada-Villa et al. \(2004\)](#) report prevalence of mental disorders and patterns of treatment use in Colombia.

A closely related Colombian study, [Marcillo-Yepey et al. \(2026\)](#), documents that the Colombia–United States free trade agreement raised property crime in agricultural municipalities, a different downstream outcome of an overlapping reform episode. Substance abuse, measured through clinical hospitalization records, remains largely unstudied in this literature, and the Colombian setting provides a particularly clean test because drug-related dynamics have historically been driven by supply-side activity that the empirical strategy can separate from the consumption-side response.

3 Background

This section provides the institutional context for the 2010 reform and for the data used to measure substance abuse. I first sketch Colombia’s path through trade liberalization, with attention to the 2010 unilateral reform that generates the identifying variation. I then describe the healthcare system that produces the administrative records used as outcomes, together with the patterns of substance use in the country.

3.1 Trade liberalization in Colombia

Colombia spent the second half of the twentieth century transitioning from import-substitution protectionism to a more open trade regime. The average nominal tariff fell from 51.9% in 1970 to 25.9% in 1981 and 6.1% in 1991, with the largest cuts concentrated in initially high-tariff sectors and a narrowing of the dispersion across tariff lines during *La Apertura* of 1991–1994 ([Garay, 1998](#); [Eslava et al., 2004](#)).⁸ From 2002 to 2007, Colombia’s tariff schedule was effectively fixed by the Andean Community’s Common

⁸Non-tariff measures (NTMs) such as import quotas are harder to monitor over time and by sector. [Goldberg and Pavcnik \(2007\)](#) report a high correlation between tariffs and NTMs in Colombia, and the most binding NTMs in the period of interest, including price bands and sanitary measures, were concentrated in agriculture and remained in place across the 2010 reform window.

External Tariff (CET). In 2008, Colombia stopped applying the CET, regaining the ability to set tariffs unilaterally (Rivera Pérez et al., 2021).

In November 2010, the incoming government of President Juan Manuel Santos issued Decree 4114, a unilateral tariff cut that applied on a most-favored-nation (MFN) basis to all trading partners (Bonilla-Mejía et al., 2026). The immediate bite of the decree was concentrated in manufacturing, whose tariffs stepped down with the new schedule in 2011; the mining cuts in *TVM* are realized mostly in 2014, when the manufacturing line steps down again alongside the Colombia–European Union FTA. The MFN reconstruction in Appendix Table A9 strips the FTA-period preferential cuts out of the exposure measure and is essentially unchanged from the AHS baseline, so the substance-abuse response is identified off the 2010 unilateral cut rather than off the FTA-period mining variation. The cut was published in late 2010 and took effect in 2011, so its bite first appears in the 2011 tariff schedule used to build *TVM*; this is also why the event-study base period is 2010s1, the last semester before the new rates applied. Two free trade agreements (FTAs) entered into force during the sample window and added further industry-specific cuts: the Colombia–United States FTA in May 2012 and the Colombia–European Union FTA in August 2013. Figure 1 Panel A shows the average effectively applied tariff falling from 12% in 2010 to 8% in 2011 and 6.3% by 2014. Panel B plots the same series separately for agriculture, manufacturing, and mining: manufacturing tariffs step down in 2011 with the unilateral reform and again in 2014, mining tariffs drop mostly in 2014, and agricultural tariffs decline gradually over the window.⁹

3.2 Healthcare system in Colombia

Identifying the effect of trade liberalization on substance abuse requires a setting where changes in substance-related health outcomes can be observed at the population level. Colombia’s healthcare system provides such a setting: near-universal insurance coverage and mandatory reporting by all health providers generate administrative records that capture hospitalizations and emergency room visits across the country. This subsection describes the institutional features that make these data a reliable

⁹The slower pace of agricultural tariff cuts reflects continuing protections: the Andean System of Price Bands kept ceiling, floor, and reference prices for 13 sensitive agricultural products, and the agricultural provisions of both FTAs were subject to long phase-in schedules.

source for measuring substance abuse rates.

In Colombia there are two types of regimes for access to health services: contributory and subsidized, covering the beneficiary's dependent family members (children and spouse). In the contributory regime, formal employees and self-employed workers pay into the social security system through payroll. The subsidized regime covers people in conditions of vulnerability, who must meet a poverty score or belong to a susceptible group. Healthcare coverage is above 95%, with 56% of the insured in the subsidized regime and 44% in the contributory regime (Camacho and Mejía, 2017).

Nearly universal access to healthcare allows for coverage of generalist and specialist medical attention at moderate cost, and in the event of an emergency those without insurance can be seen without restriction. According to the 2015 Demographic Health Survey (DHS), 64.6% of respondents with health problems sought formal medical treatment within the last 30 days, with the remainder relying on pharmacies, self-medication, or home remedies (Profamilia, 2015). Copays are modest in both regimes: income-graduated and capped at small fractions of the daily minimum wage in the contributory regime, and capped at 10% of total cost in the subsidized regime, with most services free at the point of use.¹⁰

The comparatively low cost incurred by patients reinforces this point: as a percentage of current health expenditures, out-of-pocket expenditure was 16%, lower than the average in Latin America (29.8%), middle income countries (36.4%) and upper-middle income countries (32.5%), respectively.¹¹ Broad access and low costs alleviate the concern of self-selection into healthcare based on confounders such as income, supporting inference at the aggregate level on changes in substance abuse as a response to trade liberalization. As I will discuss with further detail in Section 5, the health records I employ are instrumental in providing a reliable source for substance abuse in Colombia. The outcome variable measures healthcare use for substance-related diagnoses rather than substance-use prevalence directly; near-universal coverage and low out-of-pocket costs reduce but do not eliminate this gap, and I return to it in Section 6.3.

¹⁰A comprehensive introduction to the set of benefits, costs and regulatory laws for the Colombian health system can be found here: <https://www.minsalud.gov.co/sites/rid/Lists/BibliotecaDigital/RIDE/VP/DOA/RL/cartillas-de-aseguramiento-al-sistema-general-de-seguridad-social-en-salud.pdf>.

¹¹Figures reported for 2014. Data comes from the Global Health Expenditure Database: <https://apps.who.int/nha/database>.

Substance use in Colombia: Substance use in Colombia rose over the two decades preceding the 2010 reform. National surveys from the Dirección Nacional de Estupeficientes document increases in drug consumption across genders, ages, and socioeconomic strata between 1996 and 2013 (Camacho González et al., 2016). In the administrative records used in this paper, alcohol accounts for the largest share of substance-related hospitalizations among the working-age population and multiple-substance use is the next-most-common category (Table A1). The records capture clinical episodes of substance abuse (hospitalizations and emergency-room visits coded as substance-related), which is a margin distinct from the survey-based prevalence figures that document casual use in the general population.

4 Empirical strategy

Building on the institutional setting, I now specify the empirical model that links the 2010 reform to municipal substance-abuse rates. I begin with the baseline difference-in-differences specification and the Bartik-style measure of tariff exposure that serves as the treatment. I then introduce the dynamic event-study specification used to test the parallel-trends assumption and trace the post-reform response over time.

4.1 Baseline model

I estimate the effect of the 2010 tariff reform on substance abuse with the continuous-treatment difference-in-differences specification in equation 1, estimated by Ordinary Least Squares (OLS):

$$Y_{m,t} = \alpha + \mu_m + \lambda_t + \beta \cdot TVM_{m,y(t)} + \sum_{\tau \in X_m} \gamma_\tau (\tau \times Post_t) + \epsilon_{m,t} \quad (1)$$

The outcome $Y_{m,t}$ for municipality m in year-semester $t = \{2009s1, \dots, 2014s2\}$ is the rate of substance-abuse hospitalizations and ER visits per 100,000 population. The treatment $TVM_{m,y(t)}$ is annual; I let $y(t) \in \{2009, \dots, 2014\}$ denote the calendar year that contains year-semester t , so $TVM_{m,y(t)}$ takes the same value for both semesters of each year. The parameter of interest is β , the change in the substance-abuse rate associated with a one-unit change in $TVM_{m,y(t)}$. Time fixed effects (λ_t) and municipality

fixed effects (μ_m) capture common shocks in a given year and municipality-invariant characteristics like geographic and institutional factors, respectively. $X_m \times Post_t$ interacts pre-reform municipality characteristics with a post-2010 indicator. I follow [Belloni et al. \(2014\)](#) to select the set of controls using machine learning techniques. This allows me to be agnostic about which variables might influence the trend in substance abuse.

The set of X_m contains socioeconomic confounders: the pre-reform substance-abuse rate to rule out pre-existing trends or mean reversion; the unmet basic needs index to rule out baseline poverty that independently predicts substance abuse ([Embleton et al., 2013](#)); the share of public investment in health to rule out differences in the local policy environment around healthcare access; and the homicide rate per 100,000 to rule out exposure to violence as an independent driver of mental health and substance abuse ([Cuartas and Roy, 2019](#); [Hessel et al., 2019](#)). For precision purposes, all regressions are weighted by the working-age (18–65) population in 2009.

The treatment variable $TVM_{m,y(t)}$ is a Bartik-style tariff vulnerability measure that combines a municipality's pre-reform employment composition with the time-varying tariff structure, widely used in the literature ([Dix-Carneiro et al., 2018](#); [Dix-Carneiro and Kovak, 2017](#); [Erten et al., 2019](#)). For municipality m in year $y(t)$, $TVM_{m,y(t)}$ is a weighted average of two-digit Standard Industrial Classification (SIC, Revision 3) industry tariffs, with weights given by the 2005 employment shares in agriculture, manufacturing and mining (equation 2).¹²

$$TVM_{m,y(t)} = \sum_j \frac{Emp_{m,j,2005}}{TotalEmp_{m,2005}} \times Tariff_{j,y(t)} \quad (2)$$

In equation 2, $TotalEmp_{m,2005}$ denotes total 2005 employment in the 29 tradable two-digit SIC industries that compose the index J , so the weights $Emp_{m,j,2005}/TotalEmp_{m,2005}$ sum to one across j within each municipality, and $TVM_{m,y(t)}$ is a tradable-employment-weighted average of industry tariffs in percentage points.

Figure 2 illustrates the spatial variation in TVM at the municipality level, plotting the difference between 2010 and 2014. The color shades represent quantiles of the difference in tariff exposure, and it is evident that there is substantial variation across districts, even within the same department. The average municipality faced a reduction

¹²I follow [Kovak \(2013\)](#) and restrict the analysis to the tradable sectors. Thus, industry shares are complete and sum one for all municipalities.

of 3.56 percentage points,¹³ and this average *TVM* reduction will be employed in interpreting the regression estimates below.

The identifying variation in equation 2 has two sources: the cross-sectional dispersion in $Emp_{m,j,2005}/TotalEmp_{m,2005}$ across municipalities, and the time-series variation in $Tariff_{j,y(t)}$ across industries during 2009–2014. Under Goldsmith-Pinkham et al. (2020) (GPSS), the consistency of β rests on share exogeneity: the 2005 employment shares are pre-determined relative to the 2009–2014 sample, and β is consistent if these shares are uncorrelated with the unobserved shocks to municipal substance abuse, conditional on μ_m , λ_t and $Post_t \times X_m$.¹⁴ In Appendix B, I follow the GPSS test and show that the shares that explain most of the variation of *TVM* are not correlated with pre-existing trends of the substance abuse rates. Therefore, the identification of β is plausible under the share-exogeneity assumption.

4.2 Dynamic model

The specification in equation 1 estimates an average post-reform effect. To trace the dynamic response and to test the parallel trends assumption underlying the difference-in-differences design, I estimate the following event-study specification by OLS:

$$Y_{m,t} = \alpha + \mu_m + \lambda_t + \sum_{t \neq 2010s1} \beta_t (\overline{\Delta TVM}_m \cdot \delta_t) + \sum_{\tau \in X_m} \gamma_\tau (\tau \times Post_t) + \epsilon_{m,t} \quad (3)$$

where δ_t are year-semester indicators with 2010s1 as the base period, and $\overline{\Delta TVM}_m$ is the standardized realized tariff cut for municipality m . The realized cut is

$$\Delta TVM_m = \sum_j \frac{Emp_{m,j,2005}}{TotalEmp_{m,2005}} \times (Tariff_{j,2010} - Tariff_{j,2014}),$$

which is positive for municipalities whose pre-reform employment was concentrated in more heavily liberalized industries.

¹³Table A2 reports the summary statistics for both tariffs and for the *TVM* between 2010 and 2014 and their respective difference.

¹⁴Under Borusyak et al. (2022), identification rests on shock exogeneity: the time-varying industry tariffs $Tariff_{j,y(t)}$ are as-good-as-random across industries. Since in this setting there is only 29 SIC sectors at the two-digit level, the assumption that the identifying variation of *TVM* comes from the shocks is more limited.

A coefficient β_t therefore measures the difference in the substance-abuse rate associated with a one-standard-deviation-larger realized cut in year-semester t , relative to 2010s1. The pre-reform coefficients ($t \in \{2009s1, 2009s2\}$) test the parallel-trends assumption: under the null of common trends, they should be statistically indistinguishable from zero. The post-reform coefficients ($t \geq 2010s2$) trace the dynamic response of the outcome. The control vector $X_m \times Post_t$ and the post-double-LASSO selection procedure are the same as in equation 1, following Belloni et al. (2014). Standard errors in equations 1 and 3 are clustered at the municipality level to account for autocorrelation (Bertrand et al., 2004).

5 Data

This section describes the data sources used to estimate the specifications introduced above. The dataset combines administrative health records, the 2005 census, industry-level tariffs, household-survey labor data, and a municipality-level panel of socioeconomic and conflict characteristics.

Health records: Data on health outcomes come from *Registros Individuales de Prestadores de Salud* (RIPS). RIPS data register the reported visits, hospitalizations and ER visits at the individual level from both public and private health providers across all of municipalities in Colombia, ranging from small clinics and laboratories to big hospitals. The provision of information to the centralized system (*Sistema Integrado de la Información de la Protección Social*, SISPRO) is required by Resolution 3374 of 2000, which mandates all health providers to report all procedures, costs, length of the visit and diagnostics to the Ministry of Health.¹⁵

RIPS data record diseases based on all levels of ICD-10.¹⁶ I use municipality-based records provided by the Ministry of Health for the period 2009-2014 for the working-

¹⁵The aggregated municipality-level extracts I obtain from the Ministry of Health are coded by the municipality of the reporting provider rather than the municipality of patient residence. If exposed-municipality residents are treated in unexposed large cities (Bogotá, Medellín, Cali), the per-capita rate of their home municipality is understated and the rate of the destination city is overstated, which would attenuate the cross-municipality response on average and concentrate measured cases in cities. This cross-municipality treatment pattern is one reading of why the OLS rate response is stable to dropping the five largest cities while the PPML count response is concentrated in them (Section 6.2). The 90.86% own-municipality work share documented for the labor measure does not extend to hospital choice, since serious-care patients often travel to higher-tier facilities outside their home municipality.

¹⁶Listed by the World Health Organization, ICD-10 is the 10th revision of the International Statistical Classification of Diseases and Related Health Problems.

age population (18-65 years old), which allows me to observe health outcomes before and after the policy change. I am able to track the information on the overall number of hospitalizations and ER visits due to intoxication and harmful use of psychoactive substances (ICD-10 codes F10-F19) and all its coded sources: alcohol, hallucinogens, cannabinoids, cocaine, volatile solvents, multiple drugs, opiates, sedatives, tobacco and other stimulants.¹⁷ In all cases, I compute the rate per 100,000 population.¹⁸ Moreover, I observe both the number of distinct patients and the number of services provided for each condition. The patient rate captures the *extensive* margin of treatment, in terms of how many individuals enter care, while the service rate captures overall treatment volume. Because total services scale with both the number of patients and the services delivered per patient, the service rate also embeds the extensive change and is best read as an upper bound on a pure intensive-margin response rather than a clean measure of it. Even though there could be some risk of misreporting given that I only observe actual ER visits and hospitalizations, RIPS data have been validated and used frequently in the literature as a source of health records in Colombia (Aristizabal-Mayor and Rosselli, 2016; Camacho and Mejía, 2017; Iregui-Bohórquez et al., 2019; Ordoñez, 2020).¹⁹ Panels A.1 and A.2 of Table A1 show descriptive statistics of substance abuse by source. For the period of analysis, the average level of substance abuse in patients is 2.36 per 100,000 people and 2.87 services per 100,000 people. Abuse of alcohol, multiple drugs and hallucinogens are the most common across municipalities in terms of both patients and services provided. Figure 3 shows the rate of substance abuse for patients and services provided. The increase has been dramatic: both rates of substance abuse were three times as high in 2014 as their levels in 2009. Moreover, Appendix Figure A1 disaggregates this trend by tariff exposure, and municipalities above the median realized cut ΔTVM_m experience a sharper post-2010 rise than below-median municipalities, with the patient-rate gap widening from 2.2 to 5.2 cases per 100,000 between 2010 and 2014 and the service-rate gap widening from 2.2 to 7.0 cases per

¹⁷To save space, the exact name of some variables was shortened. See Table A3 for reference of each category definition based on ICD-10.

¹⁸All outcome variables are scaled by the municipality-level working-age population in any given year. Data come from population projections of the National Census in 2005.

¹⁹Using RIPS also represents an advantage because it is available biannually since 2009, compared with other sources like the surveys available from the National Narcotics Administration, conducted in 1992, 1996 and 2008, and the National Survey on Mental Health, conducted by the Ministry of Health in 1993, 1997, 2003 and 2015.

100,000 over the same window.

Census and tariff data: I collect municipality-level data from Colombia’s population census in 2005 using the Retrieval of Data for small areas by Microcomputer (RE-DATAM), a system supported by the population division of the Economic Commission for Latin America and the Caribbean. I calculate the initial levels of employment for the working-age population (men and women 18-65 years old) by industry at 2-digit SIC code for agriculture, manufacturing and mining sectors in each municipality.²⁰ Most of the labor force (90.86%) works in the municipality where they live, so there should not be serious concern about attenuation bias due to measurement error.

Tariff data come from the World Integrated Trade Solution (WITS), which compiles the tariff schedules WTO member countries submit to the WTO Secretariat. I use the *effectively applied* tariff (AHS), the lowest duty importers actually face: the most-favored-nation (MFN) rate that applies in the absence of a trade agreement, or a lower preferential rate under an agreement in force. The 2010 unilateral reform accounts for the AHS decline observed before 2012 (from 12% to 8% between 2010 and 2011); the Colombia–United States Free Trade Agreement (effective May 2012) and the Colombia–European Union Free Trade Agreement (effective August 2013) contributed further industry-specific cuts after 2012, taking the average AHS to 6.3% by 2014.

WITS publishes tariffs at the HS 6-digit level; I aggregate to the 2-digit SIC rev. 3 level using the official crosswalk from Colombia’s statistical bureau, *Departamento Administrativo Nacional de Estadística* (DANE), for the period 2008–2014.²¹

Employment data: For the analysis of labor outcomes, I use the *Gran Encuesta Integrada de Hogares* (GEIH), Colombia’s nationally representative continuous household survey administered by DANE. GEIH samples approximately 18,000 households per month and provides individual-level information on employment status, 4-digit SIC industry of occupation, formality, and monthly and hourly wages. GEIH covers both formal and informal workers, but its sample design supports inference only at the department level.²² I aggregate the monthly micro-data to the department–semester

²⁰I observe 29 tradable 2-digit SIC sectors (3 agriculture, 4 mining, 22 manufacturing).

²¹SIC 12 (uranium and thorium ore mining) has no WITS-AHS coverage; SIC 37 (recycling) is dropped. Mejía et al. (2018) use the same WITS-AHS series to evaluate the effect of trade liberalization on labor fluidity.

²²It covers 24 out of the 32 departments in Colombia. With this small number of clusters, the labor-mechanism regressions reported in Section 6.3 use the wild cluster bootstrap to correct the cluster-robust inference (Cameron et al., 2008); implementation details are in the relevant table and figure notes.

level for 2008–2014, summing GEIH-weighted contributions across the six months of each semester to match the time structure of the substance-abuse analysis.²³ I construct employment counts and weighted-mean log wages separately for all tradable industries, manufacturing, agriculture, and mining. These aggregates serve as the outcomes in the labor-mechanism analysis.

Other data: finally, in order to account for baseline characteristics that could explain our health outcomes, I use the ready-to-use panel with an array of municipality characteristics put together by Universidad de los Andes ([Acevedo and Bornacelly, 2014](#)). It is the source of information to construct the vector of baseline characteristics for equations 1 and 3. The same CEDE data archive also provides a municipal conflict-and-violence panel with coca-cultivation and eradication outcomes, which I use in Section 6.4 to evaluate the illicit-drug-supply mechanism.

The final dataset comprises 842 municipalities for each time period, the group of municipalities where roughly 70% of the working-age population lives.²⁴ Sample sizes vary slightly across specifications because some baseline characteristics are missing for a small number of municipalities.²⁵

6 Results

This section presents the empirical results. I begin with the baseline estimates and the heterogeneity of the response across substance categories, age groups, and sex. I then document robustness to alternative specifications, samples, and outcome transformations. The final two subsections evaluate the labor-market mechanism and examine

²³The pooled sample comprises approximately 1.5 million household–month observations over 2008–2014. Informality is proxied by the worker’s lack of contributory pension coverage, following standard definition of informality in the literature ([Maloney, 2004](#)).

²⁴The estimation sample of 842 municipalities is set at the 2005 employment data construction step: a municipality enters the sample only if DANE produced a REDATAM employment-by-SIC file for it in the 2005 census and if it had non-zero recorded employment in tradable SIC industries (agriculture, mining, manufacturing). The 256 municipalities excluded relative to the 1,098 enumerated in the 2005 census are typically very small or in conflict-affected departments where census coverage was limited, or have all recorded employment in non-tradable sectors, in which case the *TVM* cannot be constructed. RIPS itself covers more municipalities; the restriction is on the treatment, not the outcome.

²⁵The 842 figure is the no-controls baseline, the 829 figure used in the main controls-included regressions (Tables 1, 2, A9, A10, A11) drops municipalities with missing 2010 public-investment-in-health data, and the 824 figure used in the balance test (Appendix B) further drops municipalities with missing log RIPS service volume in 2009s1, a variable that enters only the augmented-pool sensitivity in Table A5 and the balance test.

two alternative mechanisms, illicit-drug supply and selective migration.

6.1 Baseline results

Table 1 reports the baseline estimates of equation 1 for total substance abuse. Columns 1–4 present versions of the same specification with controls selected by the double-selection LASSO procedure of [Belloni et al. \(2014\)](#); all include municipality and year-semester fixed effects, with standard errors clustered at the municipality level. I report patients and services separately: the rate of patients per 100,000 captures the extensive margin of treatment (how many individuals enter care), while the rate of services per 100,000 captures overall treatment intensity, though it also embeds the extensive change.

Higher values of $TVM_{m,y(t)}$ denote greater remaining tariff protection, so that trade liberalization corresponds to a fall in TVM . The coefficient on $TVM_{m,y(t)}$ is negative and statistically significant in every column, indicating that municipalities exposed to deeper tariff cuts—larger declines in TVM —had higher rates of substance abuse over the post-reform period. To gauge the magnitude implied by a tariff change of representative size, I evaluate the estimates from the most saturated specifications (Columns 2 and 4) at the population-weighted average reduction in TVM between 2010 and 2014 of 3.56 percentage points (Panel B, Table A2). A cut of this average size implies an increase of 1.51 substance-abuse patients per 100,000 and 2.21 services per 100,000, equivalent to roughly 44% and 56% of the respective pre-reform (population-weighted) means.²⁶

Table 2 decomposes the aggregate coefficient by re-estimating equation 1 with sector-specific TVM measures built separately from agriculture (Column 1), manufacturing (Column 2), and mining (Column 3) tariffs. Columns (1)–(3) report univariate sectoral regressions; column (4) includes all three sector-specific TVM s jointly. The manufacturing coefficient (Column 2) is negative and significant at the 1% level in both panels and barely changes between the univariate and joint (Column 4) specifications, pointing to manufacturing as the primary channel through which the baseline

²⁶The 25th and 75th percentiles of the population-weighted distribution of the 2010–2014 reduction in TVM differ by 4.39 percentage points. Moving from the low-exposure to the high-exposure quartile, i.e. from a smaller to a larger tariff cut, implies an increase of 1.86 patients and 2.73 services per 100,000, or 55% and 70% of the pre-reform mean, respectively.

effect in Table 1 operates. In the univariate specification, the agriculture coefficient (Column 1) is of the opposite sign and significant at the 1% level. This pattern is partly mechanical: the agricultural and manufacturing employment shares used to construct sector-specific *TVM* are strongly anti-correlated across the 842 municipalities ($\rho = -0.96$), because shares of tradable employment sum to one within each municipality. This near-collinearity means the joint specification cannot cleanly partial the two sectors apart; nonetheless, when manufacturing exposure is held constant in column (4), the agriculture coefficient shrinks by roughly one-third and is significant only at the 10% level. The mining coefficient (Column 3) is positive in both panels but not statistically distinguishable from zero, reflecting that mining employment is concentrated in a small number of municipalities and contributes limited identifying variation.

Table 3 disaggregates the total substance-abuse rate into the nine ICD-10 categories tracked in RIPS. After accounting for multiple testing through [Anderson \(2008\)](#) sharpened q -values across the family of nine outcomes within each panel, the statistically robust response is concentrated on multiple-drug use (Column 6), which survives the false-discovery correction at the 1% level ($q = 0.001$ for patients and $q = 0.005$ for services), with a weaker response for sedatives (Column 9) that survives at the 10% level ($q = 0.059$ for patients and $q = 0.055$ for services). Coefficients on alcohol (Column 1), hallucinogens (Column 2), cannabinoids (Column 3), cocaine (Column 4), volatile solvents (Column 5), opiates (Column 7), and other stimulants (Column 8) are negatively signed, consistent with the aggregate direction, but do not survive the correction. The concentration is therefore one of statistical precision rather than magnitude: rescaling the multiple-drug coefficients by the 3.56 percentage-point average tariff cut implies an increase of roughly 0.43 patients and 0.59 services per 100,000, or about 28% and 27% of the aggregate effect in Table 1. So the surviving categories do not account for the bulk of the total response, which is instead spread thinly across the remaining, individually insignificant outcomes.²⁷

Table 4 examines heterogeneity in the effect by age and sex. Columns (1)–(4) report the *TVM* coefficient when the per-100,000 case rate is restricted to four age groups (18–24, 25–34, 35–44, and 45–65); columns (5)–(6) restrict to males and females. To

²⁷The pre-reform mean rates of multiple-drug use are 0.52 patients and 0.62 services per 100,000 (Table A1), so these correspond to large percentage changes from a small base.

correct for multiple hypothesis testing across the four age groups within each panel, the table reports [Anderson \(2008\)](#) sharpened q -values. The coefficient is statistically distinguishable from zero for adults aged 25–34 and 35–44 in both panels and is not distinguishable from zero for the youngest (18–24) and oldest (45–65) groups. The coefficient is larger in magnitude and more precisely estimated for males than for females in both the patient and service rates; a test of equality between the male and female coefficients rejects at the 1% level ($p < 0.001$ for patients, $p = 0.001$ for services), consistent with the literature on gendered responses to trade liberalization in developing countries ([Ben Yahmed and Bombarda, 2020](#)).

A maintained identifying assumption is that, conditional on the controls, the dynamics of substance abuse were parallel across municipalities before the reform. Figure 4 reports event-study coefficients from the dynamic specification in equation 3, in which the standardized realized tariff cut $\widetilde{\Delta TVM}_m$ is interacted with year-semester indicators relative to a 2010s1 base period. Because this specification is written in terms of the magnitude of the tariff cut rather than the level of protection, its coefficients carry the opposite sign to the level coefficient in Table 1: a positive response to a deeper cut and a negative response to a higher level of protection are the same statement that liberalization raised substance abuse. The coefficients are also expressed in standard-deviation units of the cut, and so are not directly comparable in scale to the per-percentage-point magnitudes reported above. Panel (a) plots the response of the patient rate; panel (b) plots the response of the service rate.²⁸ The pre-2010 coefficients are statistically indistinguishable from zero: the joint test of the 2009s1 and 2009s2 coefficients yields $F(2) = 0.76$ ($p = 0.470$) for patients and $F(2) = 0.09$ ($p = 0.910$) for services, consistent with parallel pre-trends, though the single pre-reform year available limits the power of this test. Post-reform coefficients are positive throughout and grow in magnitude over time, and the joint post-treatment test (2010s2–2014s2) rejects the null of no effect at the 1% level for both outcomes ($F(9) = 7.77$ for patients, $F(9) = 9.83$ for services, $p < 0.001$ in both cases). Appendix Figure A4 reports the dynamic counterpart of the static sectoral decomposition in Table 2, with the realized cut constructed separately from manufacturing tariffs (Panels (a)–(b)) and mining tariffs (Panels (c)–(d)). Pre-2010

²⁸Appendix Figures A2 and A3 report the dynamic specification for multiple drugs and sedatives, and confirm that their post-2010 build-up follows the same gradual pattern as the aggregate, with pre-2010 coefficients statistically indistinguishable from zero and joint post-treatment F -tests significant at the 1% level for both patients and services.

coefficients are statistically indistinguishable from zero in all four panels ($F(2)$ ranges from 0.08 to 1.96, all $p > 0.14$). The manufacturing-specific event study mirrors the aggregate post-2010 buildup, with the joint post-treatment test rejecting at the 1% level for both outcomes ($F(9) = 8.92$ for patients, $F(9) = 10.37$ for services); the mining-specific event study shows no consistent post-2010 response, consistent with mining's limited identifying variation in the panel. The response of substance abuse to the 2010 reform built up gradually, with visible effects emerging within two to three years of the tariff cut.

6.2 Robustness

In this section I run a series of robustness checks to validate the consistency and sensitivity of my main estimates (Table 1, Columns 2 and 4).

Specification robustness: Appendix Table A4 reports three exercises that add or modify the set of controls in the baseline specification. Column (2) adds the employment-weighted tariff on Colombia's exports to its main trading partners, addressing the concern that simultaneous changes in foreign tariffs on Colombian exports during the sample period confound the unilateral import-side reform (Erten and Keskin, 2021).²⁹ Column (3) adds the muni-year input tariff constructed following Edmonds et al. (2010), addressing the concern that the reform also lowered tariffs on imported intermediate inputs, which would benefit Colombian producers in import-affected industries and partially offset the baseline effect.³⁰ Column (4) adds department-by-year fixed effects to the baseline, absorbing time-varying department-level shocks such as regional macroeconomic conditions that could correlate with the spatial pattern of *TVM* exposure.³¹ In all three exercises the *TVM* coefficient is similar to the baseline for both patients and services, and the inclusion of department-by-year fixed effects strengthens the estimate rather than absorbing it. Appendix Table A5 ad-

²⁹The U.S.–Colombia Free Trade Agreement took effect in May 2012 and the E.U.–Colombia Free Trade Agreement was provisionally applied in August 2013, both of which lowered foreign tariffs on Colombian exports. The standard error on *TVM* rises in column (2) because this partner-tariff measure is mechanically similar to *TVM* by construction (both are 2005-employment-share-weighted averages of tariffs), producing a within-municipality and year-semester correlation of about 0.59; the point estimate is essentially unchanged.

³⁰The input tariff for each output industry i is the cost-share-weighted average tariff on industries that supply inputs to i , computed from the 2005 Colombian input–output matrix and aggregated to the municipality level using 2005 employment shares.

³¹Colombia is comprised of 32 departments, an administrative unit similar to states.

dresses a closely related concern by re-running the post-double-LASSO selection step under an extended candidate pool that adds log RIPS service volume in 2009s1 and log 2009 working-age population to the four-variable baseline pool; the *TVM* coefficient is attenuated but remains negative and statistically significant at the 5% level for both outcomes, indicating that the result is robust to the choice of candidate pool.

Alternative samples: Appendix Table A6 reports the baseline specification on two alternative samples. Column (2) restricts to the 694 municipalities with no historical coca cultivation in 2005; that exercise speaks to the illicit-drug-supply mechanism and is discussed in Section 6.4. Column (3) trims municipalities below the population-weighted first and above the ninety-ninth percentile of the realized tariff cut ΔTVM_m , addressing the concern that a small number of extreme-exposure municipalities is driving the result; the *TVM* coefficient is slightly larger in magnitude than the baseline for both patients and services.

Outcome distribution: Appendix Table A7 addresses concerns about the zero-inflated and right-skewed outcome distribution. In small municipalities, a single substance-abuse case can produce a large per-100,000 rate, so OLS estimated on rates can be sensitive to small-cell outliers. Column (1) reproduces the baseline OLS rate specification for reference. Column (2) replaces the dependent variable with the log of one plus the case rate, dampening the influence of high-rate observations. Column (3) restricts the sample to the 421 municipalities with above-median 2009 working-age population, reducing the contribution of small-cell outliers. Columns (4) and (5) estimate a pseudo-maximum-likelihood Poisson (PPML) regression on the case count with the log of 2009 working-age population as an offset (Silva and Tenreyro, 2006), with the coefficient displayed as an incidence-rate-ratio (IRR); column (4) uses the full sample and column (5) restricts to the above-median-population sample. Both PPML columns use the identical *TVM* treatment in the same protection-level orientation as the OLS columns, the same municipality and year-semester fixed effects, and the same clustering at the municipality level. Because the PPML specification includes municipality fixed effects, municipalities with zero counts in every semester are dropped by separation (510 of 842 baseline municipalities are retained in column 4); this is the fixed-effects separation phenomenon, not a property of the Poisson likelihood itself. The OLS coefficient is negative and statistically significant across all three OLS columns

for both patients and services, so the relationship documented in Table 1 is not driven by small-municipality outliers or by the per-100,000 rate transformation. The PPML incidence-rate-ratios in columns (4) and (5) exceed one, of opposite sign to the OLS columns. Column (6) re-estimates the OLS rate regression on the same 510 municipalities retained by the PPML estimator: the OLS coefficient remains negative and statistically significant at the 1% level for both patients and services. The point estimate attenuates by about a quarter relative to the full-sample baseline in column (1), so dropping the 332 zero-count municipalities does shift the OLS magnitude even though the sign and statistical significance are unchanged; this attenuation is itself consistent with the heterogeneity documented in columns (7) and (8). The sign and significance of the OLS rate response are therefore not driven by sample composition, but the magnitude is not invariant to it. To localize the source of the disagreement, columns (7) and (8) drop the five municipalities with the largest total recorded case counts (Bogotá, Medellín, Cali, Bucaramanga, and Cartagena, which together account for roughly two-thirds of recorded substance-abuse cases in the panel)³² and re-estimate the OLS rate specification (Column 7) and the PPML specification (Column 8). The OLS point estimate is essentially unchanged (the patient coefficient moves from -0.525 to -0.489 and the service coefficient from -0.715 to -0.664), so the per-capita rate response is not a phenomenon of the five largest cities. The PPML point estimate moves a lot: the patients IRR drops from 1.307 to 1.146 (the excess over one falls by roughly half, from 0.307 to 0.146) and the services IRR drops from 1.365 to 1.146 (the excess falls by about 60%, from 0.365 to 0.146), and both lose statistical significance. Both estimators lose precision when those five cities are dropped, since the five cities carry enormous population weight; the clean asymmetry is in the point estimate, not in precision. Reading the two columns together, the OLS rate response is stable to dropping the largest cities while the PPML positive association is not, which says that the rate result is a broad-municipality phenomenon and that the count-based association is concentrated in a small number of high-case-volume cities whose population-weighted case-count response to *TVM* runs the opposite way. PPML weights the gradient by the predicted case count, putting most of the leverage on those few cities, which is why the count and rate specifications can disagree on identical data. The rate and count specifica-

³²The ranking is by total RIPS substance-abuse cases over 2009–2014, not by population. Barranquilla, Colombia’s fourth-largest city by population, falls below the top five on this case-count ranking.

tions answer different questions: an average per-capita absolute change in the case rate (OLS) versus an average proportional change in case counts weighted toward the largest cities (PPML). I retain the OLS rate specification as the body benchmark because the per-capita response of the working-age population is the parameter the paper is asking about; the PPML reversal is reported as a localized heterogeneity finding rather than as a refutation of the rate result.

Bartik sensitivity: Appendix Table A8 rebuilds the *TVM* treatment after sequentially dropping the industries with the highest Rotemberg weights in the [Goldsmith-Pinkham et al. \(2020\)](#) decomposition, and re-estimates the main specification with the resulting leave-one-out treatments. The first row reproduces the baseline using all 29 tradable industries; each subsequent row drops the next industry in order of Rotemberg weight (wearing apparel, textiles, food products and beverages, fabricated metal products, and publishing and printing). The *TVM* coefficient is stable in sign and grows substantially in magnitude as the highest-weighted industries are removed, nearly doubling for the patient rate and more than doubling for the service rate when all five top-Rotemberg-weight industries are dropped. The baseline is therefore not driven by a small number of high-Rotemberg-weight industries; if anything, those industries pull the estimate toward zero, so the baseline is conservative.

Unilateral tariff variation: Appendix Table A9 reconstructs *TVM* from the most-favored-nation (MFN) applied tariff in place of the effectively-applied (AHS) tariff used in the baseline. The MFN tariff is the rate applied in the absence of preferential agreements, so an MFN-based *TVM* strips out the cuts granted under the Colombia–United States free trade agreement (effective May 2012) and the Colombia–European Union free trade agreement (effective August 2013), which entered into force part-way through the sample period. The purpose of this exercise is to ask whether the substance-abuse response is identified off the 2010 unilateral cut alone, or whether the contemporaneous FTAs are also contributing to the estimate.³³ Columns (1) and (3) reproduce the AHS-based estimate on the matched sample, and Columns (2) and (4) report the MFN-based estimate. The MFN coefficient is statistically significant at the 1% level and quantitatively close to the AHS coefficient for both patients and services,

³³Colombia’s MFN rate is not purely unilateral, as the CAN (Andean Community) and Mercosur preferences predate the sample period and also lie below MFN; the concern addressed here is the post-2010 FTA contribution, since those agreements are the ones contemporaneous with the panel.

so the substance-abuse response is identified off the 2010 unilateral cut rather than off the FTA-period variation. The two estimates are close because the cross-municipality correlation between AHS and MFN exposure is 0.99: the FTAs introduced a level shift in industry tariffs without much reordering of municipalities by exposure. The MFN check therefore rules out the FTAs as the source of the result, but it is not a fully independent identification strategy from the baseline. The dynamic version of the MFN-based specification is reported in Appendix Figure A5, which plots the event-study coefficients on $\widehat{\Delta TVM}_m$ reconstructed from MFN tariffs: the pre-2010 coefficients are statistically indistinguishable from zero ($F(2) = 0.95, p = 0.39$ for patients; $F(2) = 0.07, p = 0.94$ for services), and the joint post-treatment tests reject the null at the 1% level ($F(9) = 8.80$ for patients; $F(9) = 10.42$ for services), so the parallel-trends assumption and the post-reform response are preserved when the FTA contribution is stripped out of the exposure measure.

Reporting-quality placebo: A final concern is that the documented effect could reflect differential changes in RIPS reporting quality across municipalities rather than a genuine increase in substance abuse. I re-estimate the main specification on five acute-condition outcomes that share the RIPS reporting channel. Appendix Table A10 reports the TVM coefficient when the outcome is replaced with the per-100,000 rate of patients and services in each of five ICD-10 chapters: certain infectious and parasitic diseases (Column 1), diseases of the circulatory system (Column 2), diseases of the digestive system (Column 3), congenital malformations (Column 4), and external causes including accidents and injuries (Column 5). The pattern across the five chapters cuts against a reporting-quality story in two ways. First, the placebo coefficient is statistically indistinguishable from zero for the infectious, digestive, and congenital outcomes; if differential improvements in RIPS reporting in high-exposure municipalities were inflating the substance-abuse counts, they should inflate these counts as well. Second, the coefficient on external causes is significant at the 10% level and of opposite sign to the substance-abuse result, which is hard to reconcile with a reporting-quality channel that improves the recording of acute conditions broadly. The circulatory-disease coefficient is significant and of the same sign as substance abuse; this is plausibly downstream of the substance-abuse increase, given the established cardiovascular effects of alcohol (George and Figueredo, 2011) and cocaine (Lange and Hillis, 2001), rather than

a reporting artifact, although I do not attempt to disentangle the two channels in this paper.

Taken together, the exercises in this section indicate that the documented relationship between tariff cuts and substance abuse is not an artifact of the specific choice of controls, sample, outcome distribution, treatment construction, or candidate pool for the post-LASSO control selection. The TVM coefficient remains negative across the modifications considered, and the placebo outcomes show no response, supporting a causal interpretation of the baseline estimates.

6.3 Labor market adjustments

Trade liberalization can affect substance abuse through several pathways, but the one most discussed in the literature is labor-market disruption (Adda and Fawaz, 2020; Pierce and Schott, 2020). Workers displaced from import-competing sectors face job loss, lower wages, and uncertainty, and the public-health literature documents that this configuration raises the risk of stress-related health behaviors including alcohol and drug use (Colantone et al., 2019; Ruhm, 2015). I assess whether this channel operates in the 2010 Colombian reform using GEIH labor outcomes for the prime-age (25–44) working population, aggregated to the department–semester level for 2008s1–2014s2. The 25–44 age band is the one in which the substance-abuse response is concentrated: Table 4 shows that the TVM coefficient is statistically significant for adults aged 25–34 and 35–44 but not for the younger (18–24) and older (45–65) groups, so the labor outcomes most relevant to the mechanism are those of workers in the same age range. The choice of department-level aggregation is dictated by the GEIH sampling design, which supports inference only at the department level.

Table 5 reports four panels by tradable-sector group (all tradables, manufacturing, agriculture, and mining), with five outcomes per panel: log total employment, log informal employment, log formal employment, log monthly wage, and log hourly wage. The treatment is the department-level $TVM_{d,t}$ constructed from 2005 department-level employment shares.³⁴

The total-employment and informal-employment coefficients (Columns 1 and 2) in Panels A and B are not statistically distinguishable from zero, and neither are

³⁴See Appendix C for a definition of $TVM_{d,t}$.

the wage coefficients (Columns 4 and 5). The one significant result is on formal employment. For all tradable sectors (Panel A) and for manufacturing alone (Panel B), the *TVM* coefficient on log formal employment (Column 3) is positive, with point estimates of 0.040 in both panels (recall that *TVM* is in protection-level orientation, so a positive coefficient says that more tariff protection sustains employment and a tariff cut reduces it). With 24 department clusters, the asymptotic cluster-robust standard error may overstate significance (Cameron et al., 2008), so Table 5 also reports the wild cluster bootstrap p -value in square brackets below the cluster-robust standard error. Under this small-cluster correction the formal-employment estimate is significant at the 5% level in Panel A ($p = 0.043$) and at the 10% level in Panel B ($p = 0.069$).³⁵ Evaluating Panel A at the 2.81 percentage-point population-weighted average tariff cut for the department-level *TVM* (Panel C of Table A2; this is the department-level average and is smaller than the 3.56 percentage-point municipality-level cut used in the substance-abuse magnitude calculation in Section 6.1), this implies a contraction in formal tradable-sector employment of $0.040 \times -2.81 \approx -0.112$ log points, i.e. about a 10.6% contraction ($e^{-0.112} - 1 \approx -10.6\%$), over the 2009–2014 window. Since informal employment does not move, the formal-sector loss is not offset by informal-sector absorption within the same sector; since wages do not move either, the adjustment operates along the extensive (worker count) margin rather than the intensive (wage) margin in the short run (Dix-Carneiro, 2014).

Panels C (agriculture) and D (mining) are less informative for the substance-abuse channel. In agriculture (Panel C), the coefficient on log informal employment (Column 2) is positive and significant at the 5% level under the wild cluster bootstrap ($p = 0.018$); formal employment (Column 3) is marginally significant ($p = 0.095$); the wage coefficients (Columns 4 and 5) are not distinguishable from zero under the small-cluster correction even though the hourly-wage coefficient is significant under the asymptotic cluster-robust standard error. Compared with manufacturing, agriculture saw little change in tariffs over the 2009–2014 window (Panel B of Figure 1), so this panel is not where the labor-market shock from the reform is concentrated. In mining (Panel D), the monthly-wage and hourly-wage coefficients (Columns 4 and 5) are negative and statistically significant under the cluster-robust standard error but are not

³⁵Wild cluster bootstrap implemented with Stata's `wildbootstrap` command using 9,999 Rademacher-weight replications.

distinguishable from zero under the wild cluster bootstrap ($p = 0.128$ and $p = 0.265$ respectively), and no employment effect is detected.

Figure 5 reports the dynamic version of the formal-employment result. Panel (a) plots event-study coefficients from a regression of log prime-age formal employment in all tradables on the standardized realized cut $\widetilde{\Delta TVM_d}$ interacted with year-semester indicators (2010s1 base period); Panel (b) does the same for manufacturing. Both panels show 95% wild cluster bootstrap confidence intervals (Rademacher weights, 9,999 replications); Appendix Figure A6 shows the same coefficients with asymptotic cluster-robust intervals for comparison. The pre-2010 coefficients are statistically indistinguishable from zero in both panels: the joint pre-trend wild cluster bootstrap p -values are 0.457 for all tradables and 0.409 for manufacturing.³⁶ Post-treatment coefficients are negative throughout and grow in magnitude over time. The cluster-robust joint post-treatment tests reject at the 1% level ($F(9) = 5.45$ for all tradables; $F(9) = 3.93$ for manufacturing), but under the wild cluster bootstrap the joint significance is sensitive to the small-cluster correction ($p = 0.132$ for all tradables; $p = 0.208$ for manufacturing). The dynamic shape, with coefficients building up gradually over two to three years, parallels the gradual buildup of the substance-abuse response documented in Section 6.1. Because the event-study joint test does not survive the small-cluster correction, the inferential weight of the labor mechanism rests on the static specification reported in Table 5.

Consistent with this mechanism, the substance-abuse response is concentrated in the population segment most exposed to the formal-employment shock. Appendix Table A11 splits the substance-abuse outcome by health-insurance regime and re-estimates the baseline specification separately for the contributory regime, which enrolls formal employees and self-employed contributors, and the subsidized regime, which covers people in conditions of vulnerability outside formal employment. The TVM coefficient is negative and statistically significant at the 1% level in the contributory regime for both patients and services and is statistically indistinguishable from zero in the subsidized regime, so the substance-abuse increase concentrates in the

³⁶The joint pre-trend and post-treatment hypothesis tests use Roodman's boot test implementation of the wild cluster bootstrap (Roodman et al., 2019), since Stata's `wildbootstrap` command tests each linear restriction separately rather than jointly. Both commands implement the same Rademacher-weight wild cluster bootstrap with 9,999 replications, and give numerically close per-coefficient p -values where they overlap.

population to which the labor-market mechanism applies.

The regime label is recorded at the hospitalization, not at the time of any preceding displacement. A worker who loses a formal job exits the contributory regime, but movement into the subsidized regime is not automatic: eligibility is means-tested through the *Sisbén* classification rather than tied to employment status, so a recently displaced formal worker with above-threshold household assets or income does not immediately qualify.³⁷ The displacement-into-subsidized reading therefore overstates the fluidity of the regime border. Even so, I test for composition movement directly: I re-estimate the baseline specification with the contributory share of the substance-abuse patient cohort itself, $pat_{contr}/(pat_{contr} + pat_{subs})$, as the dependent variable. Appendix Table A12 reports the result: the *TVM* coefficient is -0.022 ($p = 0.071$) for patients and -0.023 ($p = 0.074$) for services on a pre-reform mean of 0.47.³⁸ The negative sign goes in the opposite direction to what a contributory-to-subsidized re-enrollment story would predict, and the magnitude is small (about 2 percentage points off a 47% base), so regime composition within the patient cohort is roughly stable with respect to *TVM*.³⁹

Together, Table 5 and Figure 5 establish three facts relevant to the substance-abuse mechanism. First, the 2010 reform reduced formal employment in the most exposed tradable sectors (Panel A); the same point estimate appears in the manufacturing subsample (Panel B), although manufacturing alone is only marginally significant under the small-cluster correction. Second, the contraction is not offset by an expansion in informal employment within the same tradable sector, so workers displaced from formal jobs did not find informal substitutes in the short run. Third, wages did not adjust meaningfully in the short run, so the labor-market response operated through job loss rather than wage cuts. These three features map onto the conditions identified in the stress-and-substance-abuse literature: formal job loss with limited informal

³⁷*Sisbén* (*Sistema de Identificación de Potenciales Beneficiarios de Programas Sociales*) is the household survey used by the Colombian government to target social programs, including subsidized health enrollment. Re-classification is administrative and typically takes months, and households between regimes can remain uninsured until they re-enroll through an employer or qualify for a subsidy.

³⁸The sample is restricted to municipality–semesters with at least one F10–F19 case, since the share is otherwise undefined ($N = 1,507$ across 336 municipalities). The specification otherwise matches the body regressions: municipality and year-semester fixed effects, the four-variable BCH control pool interacted with *Post*, weighted by 2009 working-age population, with standard errors clustered at the municipality level.

³⁹This is a regime-share test on the patients I observe rather than on the working-age population, so it does not rule out re-enrollment effects in the broader population. It does say that, conditional on appearing in the data, the regime label is not shifting in the direction that would undermine the split.

absorption and rigid wages is the configuration shown to elevate alcohol and drug use after labor-market shocks (Ruhm, 2015), and is consistent with the evidence in Pierce and Schott (2020) and Adda and Fawaz (2020) that locally-concentrated trade shocks generate increases in stress-related health outcomes through labor-market disruption.

6.4 Additional mechanisms

Beyond the labor channel, two alternative mechanisms could plausibly account for the substance-abuse response: changes in the local supply of illicit drugs and selective migration. I assess each in turn.

Local drug supply: Substance-abuse response could reflect changes in the local supply of illicit drugs rather than labor-market distress. I operationalize local drug supply through coca cultivation: Colombia is by a wide margin the world's largest producer of coca (UNODC, 2021), and coca is the primary domestically-grown input to the country's illicit-drug economy. If liberalization shifted displaced workers into coca farming, or if the broader rural economy generated more coca production in more-exposed municipalities, the substance-abuse increase could reflect availability rather than demand. Appendix Table A13 estimates the Poisson version of the baseline specification on five coca-cultivation outcomes from the CEDE municipal conflict-and-violence panel: net hectares of coca cultivated at the end of the calendar year (Column 1, i.e. planted area surviving eradication), the number of coca plots (Column 2), hectares of aerial coca eradication (Column 3), hectares of manual coca eradication (Column 4), and total coca hectares (Column 5, gross planted area before eradication; the sum of Columns 1, 3, and 4). The coefficients on coca area (Column 1) and coca plots (Column 2) are positive and statistically significant at the 1% level once the four baseline controls are included. Recalling that *TVM* measures the level of tariff protection, a positive coefficient says that more-protected municipalities have more coca activity. Equivalently, liberalization is associated with *less* coca cultivation in the affected municipalities, which is the opposite direction to what a supply-side mechanism for the substance-abuse result would require: a labor-displacement-into-coca story would predict *TVM* to enter negatively in coca-cultivation regressions. The total-coca-hectares estimate (Column 5) is close to zero and not significant, and the two eradication outcomes (Columns 3 and 4) do not produce a robust pattern across the no-controls and controls-included specifications.

A complementary sample-restriction check reported in Column (2) of Appendix Table A6 re-estimates the baseline substance-abuse specification on the 694 municipalities with no historical coca cultivation in 2005, dropping the municipalities in which the drug-supply channel could mechanically operate. The *TVM* coefficient is essentially unchanged for both the patient and service rates, so the substance-abuse response is not concentrated in coca-cultivating municipalities. The combined message is that the substance-abuse effect is not coming through expanded local cocaine availability.

Selective migration: If the 2010 reform pushed working-age residents to relocate away from more-exposed municipalities, the substance-abuse result could reflect a change in the composition of the working-age population rather than a labor-disruption response of incumbent workers. Appendix Table A14 addresses this concern by re-estimating the baseline specification with the log working-age (18–65) population at the municipality–year level as the dependent variable, taken from the DANE municipal population projections. The *TVM* coefficient is small and not statistically distinguishable from zero in either the no-controls specification (Column 1) or the controls-included specification (Column 2). This rules out a net population response to the tariff cut and therefore a composition change that would show up as a change in headcount; it does not by itself exclude offsetting selective in- and out-migration that leaves total working-age population unchanged but shifts its composition. The substance-abuse increase therefore cannot reflect a net headcount change in more-exposed areas, though the migration test is silent on compositional turnover.

7 Conclusions

This paper studies the effect of the 2010 Colombian tariff reform on substance abuse among the working-age population. Using administrative health records and a continuous-treatment difference-in-differences design, I find that municipalities more exposed to the tariff cut experienced larger increases in substance-abuse hospitalizations and emergency-room visits relative to less-exposed municipalities. Multiple-drug use is the only category whose response remains individually robust to multiple-testing correction, with a weaker effect on sedatives; the aggregate effect is statistically present for adults aged 25–44 and is larger and more precisely estimated for males than for

females.

The evidence is consistent with a labor-market channel. The reform reduced formal employment in the most exposed tradable sectors, with the adjustment running through job loss rather than wages: informal employment within the same sector did not expand, and tradable-sector wages did not move meaningfully over the post-reform window. The static labor estimate is significant at conventional levels under the wild cluster bootstrap that the 24-department setting calls for, but its dynamic counterpart is not, so the channel is best read as suggestive rather than tested as a mediation pathway. Splitting the substance-abuse outcome by health-insurance regime supports this reading: the response concentrates in the contributory regime, which covers formal employees and self-employed contributors, and is indistinguishable from zero in the subsidized regime, which serves people outside formal employment. The contributory-regime concentration locates the substance-abuse response in the population segment most exposed to the formal-employment shock. Two alternative mechanisms are not supported by the data: tariff exposure does not predict coca cultivation or eradication, and the working-age population does not shift across municipalities.

Several limitations bound the interpretation. The administrative records capture clinical episodes rather than population prevalence, missing the population that never seeks treatment. The lower-bound reading of the estimates requires non-differential under-reporting with respect to *TVM*, or, more weakly, that the labor-market shock does not lower treatment-seeking conditional on use. Loss of contributory coverage following formal-employment displacement could violate this in the opposite direction, while freed time from job loss could violate it in the same direction as the documented response; the relative magnitudes are not pinned down in the data, so the lower-bound language should be understood as conditional on these care-seeking margins. The estimates are local exposure effects on the working-age population in each municipality rather than individual treatment effects on workers in liberalized industries. The labor-mechanism analysis is identified at the department–semester level because the GEIH sampling design does not support municipality-level inference, which limits the precision of the mechanism estimates relative to the substance-abuse estimates.

The findings carry implications for policy in settings where trade reforms are paired with limited social protection. Mental health and substance abuse treatment capacity

in regions vulnerable to trade shocks is a margin where the Colombian health system has documented gaps in prevalence and treatment uptake ([Posada-Villa et al., 2004](#)). Colombia took steps on this margin during the sample window itself: Law 1566 of 2012 established a national framework for addiction treatment, Law 1616 of 2013 expanded mental-health service provision, and the Citizen Security Law of 2011 reshaped the legal treatment of small-quantity drug possession. The national content of these reforms is absorbed by the year fixed effects in the baseline specification; the residual concern is regional variation in implementation. Laws 1566 and 1616 in particular expanded the demand-side infrastructure on which such a response would operate. The formal-employment response combined with the absence of informal absorption points to active labor-market programs aimed at displaced formal workers as a complementary policy lever. More broadly, the consumption-side health response documented here adds substance-abuse hospitalization to the set of downstream social costs of trade liberalization that need to be weighed against the productivity and price gains that drove the original reform. This calculus is sharper in low- and middle-income countries, where the social-protection infrastructure for absorbing labor-market shocks is thinner than in the developed-country settings where most of the existing evidence has been produced.

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Figures and tables

Tables

Table 1: Effect of trade liberalization (*TVM*) on substance-abuse case rates (2009s1–2014s2)

	(1) Patients	(2) Patients	(3) Services	(4) Services
TVM (tradable)	-0.525*** (0.081)	-0.424*** (0.130)	-0.715*** (0.108)	-0.622*** (0.181)
Mean of dep. var. (pre-reform)	3.39	3.40	3.91	3.93
Observations	10,104	9,948	10,104	9,948
Municipalities	842	829	842	829
Municipality FE	Yes	Yes	Yes	Yes
Year-semester FE	Yes	Yes	Yes	Yes
Controls × post	No	Yes (BCH)	No	Yes (BCH)
Selected controls (BCH union)	–	4 / 4	–	4 / 4

Note: The dependent variable is the substance-abuse case rate per 100,000 working-age population at the municipality–semester level, computed from RIPS administrative records of hospitalizations and emergency-room visits with ICD-10 F10–F19 primary diagnoses. Columns (1)–(2) use the patient rate (unique patients); columns (3)–(4) use the service rate (services delivered). The treatment is the time-varying Tariff Vulnerability Measure (*TVM*), constructed as the 2005-census employment-share-weighted average of WITS-WTO effectively applied tariffs in each tradable 2-digit SIC industry. All regressions are estimated by OLS with municipality and year-semester fixed effects, weighted by 2009 working-age (18–65) population, with standard errors clustered at the municipality level. Columns (2) and (4) add baseline controls interacted with a post-reform indicator ($\mathbb{1}[\text{semt} \geq 2010\text{s}2]$); the controls are selected via the post-double-LASSO procedure of [Belloni et al. \(2014\)](#) from the candidate pool: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, homicide rate per 100,000. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 2: Sector-specific *TVM* exposure and substance-abuse case rates

	Agriculture (1)	Manufacturing (2)	Mining (3)	Joint (4)
<i>Panel A: Patient rate (per 100,000)</i>				
TVM (agriculture)	1.516*** (0.523)			0.994* (0.530)
TVM (manufacturing)		-0.473*** (0.130)		-0.419*** (0.132)
TVM (mining)			1.639 (1.224)	1.413 (1.197)
<i>Panel B: Service rate (per 100,000)</i>				
TVM (agriculture)	2.278*** (0.760)			1.534* (0.804)
TVM (manufacturing)		-0.689*** (0.180)		-0.608*** (0.184)
TVM (mining)			1.073 (1.936)	0.738 (1.816)
Mean of dep. var. (pre-reform)	3.93	3.93	3.93	3.93
Observations	9,948	9,948	9,948	9,948
Municipalities	829	829	829	829
Municipality FE	Yes	Yes	Yes	Yes
Year-semester FE	Yes	Yes	Yes	Yes
Specification	Univariate	Univariate	Univariate	Joint

Note: Columns (1)–(3) report univariate sectoral regressions in which only the indicated sector-specific *TVM* enters the equation; column (4) includes agriculture, manufacturing, and mining *TVM* jointly in the same regression. The *TVM* measure for sector *s* is the 2005 employment-share-weighted average tariff over the 2-digit SIC industries belonging to *s*. Panel A uses the patient case rate per 100,000; Panel B uses the service rate. The four columns share an identical four-variable baseline control pool (pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000, each interacted with $Post_t$), included directly so that all four columns are estimated on the same sample of 829 municipalities. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age (18–65) population, and report standard errors clustered at the municipality level (in parentheses). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Effect of trade liberalization on the rate of substance abuse by substance source

	(1) Alcohol	(2) Hallucinogens	(3) Cannabinoids	(4) Cocaine	(5) Volatile solvents	(6) Multiple drugs	(7) Opiates	(8) Other stimulants	(9) Sedatives
<i>Panel A. Patients</i>									
TVM (tradable)	-0.070 (0.086)	-0.091 (0.057)	-0.012 (0.016)	-0.029 (0.030)	-0.010 (0.013)	-0.122*** (0.031)	-0.026 (0.019)	-0.008 (0.008)	-0.076** (0.030)
Sharpened q -value	0.431	0.336	0.431	0.431	0.431	0.001	0.372	0.431	0.059
Observations	9,948	9,948	9,948	9,948	9,948	9,948	9,948	9,948	9,948
Municipalities	829	829	829	829	829	829	829	829	829
Mean of dep. var. (pre-reform)	1.325	0.540	0.081	0.160	0.187	0.519	0.097	0.084	0.488
<i>Panel B. Services</i>									
TVM (tradable)	-0.100 (0.105)	-0.159 (0.103)	-0.026 (0.016)	-0.028 (0.031)	-0.015 (0.020)	-0.166*** (0.048)	-0.032 (0.023)	-0.011 (0.011)	-0.092** (0.037)
Sharpened q -value	0.407	0.276	0.276	0.407	0.461	0.005	0.285	0.407	0.055
Observations	9,948	9,948	9,948	9,948	9,948	9,948	9,948	9,948	9,948
Municipalities	829	829	829	829	829	829	829	829	829
Mean of dep. var. (pre-reform)	1.478	0.668	0.084	0.184	0.213	0.617	0.119	0.097	0.527

Note: Each column reports the coefficient on the time-varying tradable TVM from a separate regression with the per-100,000 case rate of the indicated substance as the dependent variable. Panel A uses the patient rate; Panel B uses the service rate. The third row in each panel reports Anderson (2008) sharpened q -values controlling the false-discovery rate over the family of nine substance outcomes within the panel. Each regression adds baseline controls interacted with a post-reform indicator, selected via the Belloni et al. (2014) post-double-LASSO procedure from a four-variable candidate pool: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age (18–65) population, and report standard errors clustered at the municipality level (in parentheses). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Effect of trade liberalization on the rate of substance abuse by age and sex

	(1) Age 18–24	(2) Age 25–34	(3) Age 35–44	(4) Age 45+	(5) Male	(6) Female
<i>Panel A. Patients</i>						
TVM (tradable)	0.008 (0.038)	-0.668** (0.292)	-0.748*** (0.228)	-0.222 (0.165)	-0.653*** (0.198)	-0.234** (0.102)
Male = Female (<i>p</i> -value)					0.000	
Sharpened <i>q</i> -value	0.833	0.034	0.003	0.216	0.003	0.034
Observations	9,948	9,948	9,948	9,948	9,948	9,948
Municipalities	829	829	829	829	829	829
Mean of dep. var. (pre-reform)	0.394	1.337	7.002	4.477	4.349	2.556
<i>Panel B. Services</i>						
TVM (tradable)	-0.020 (0.047)	-1.093*** (0.346)	-0.881*** (0.291)	-0.391 (0.246)	-0.901*** (0.278)	-0.390*** (0.134)
Male = Female (<i>p</i> -value)					0.001	
Sharpened <i>q</i> -value	0.669	0.005	0.005	0.135	0.005	0.006
Observations	9,948	9,948	9,948	9,948	9,948	9,948
Municipalities	829	829	829	829	829	829
Mean of dep. var. (pre-reform)	0.411	1.535	8.055	5.232	4.942	3.018

Note: Each column reports the coefficient on the time-varying tradable TVM from a separate regression with the per-100,000 case rate of the indicated demographic group as the dependent variable. Panel A uses the patient rate; Panel B uses the service rate. Each regression adds baseline controls interacted with a post-reform indicator, selected via the [Belloni et al. \(2014\)](#) post-double-LASSO procedure from a four-variable candidate pool: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age (18–65) population, and report standard errors clustered at the municipality level (in parentheses). The “Male = Female (*p*-value)” row beneath the Male column reports the *p*-value from a test that the TVM coefficient is identical across the male and female outcomes; this is computed from a stacked regression that interacts TVM with a male indicator and includes municipality-by-sex and semester-by-sex fixed effects. The “Sharpened *q*-value” row reports [Anderson \(2008\)](#) BH/FDR adjusted *q*-values over the family of six demographic outcomes within each panel. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

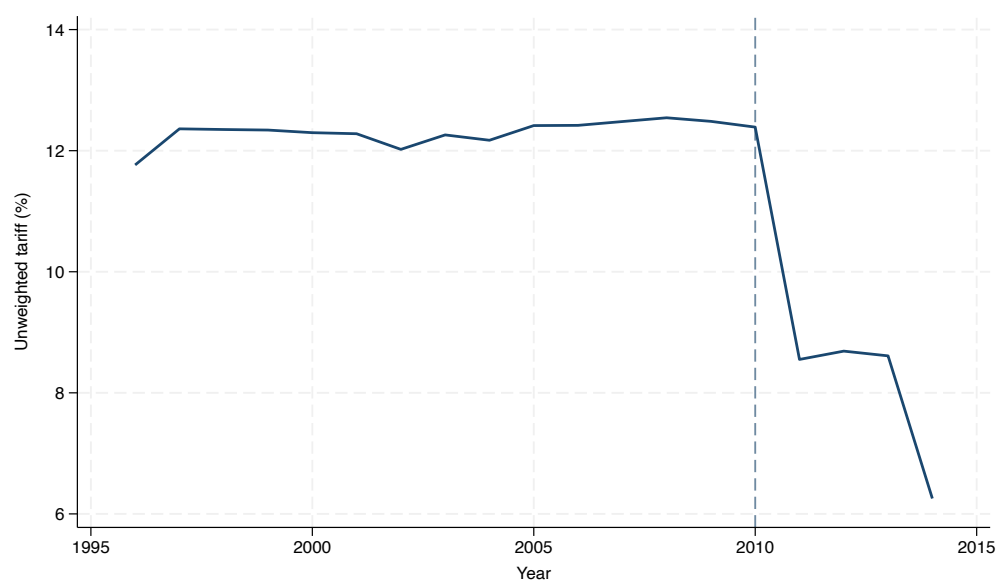
Table 5: Effect of trade liberalization on prime-age (25–44) labor outcomes at the departmental level

	ln(Total emp) (1)	ln(Informal emp) (2)	ln(Formal emp) (3)	ln(Monthly wage) (4)	ln(Hourly wage) (5)
<i>Panel A: All tradable sectors (SIC 1–36)</i>					
TVM	-0.002 (0.007) [0.813]	-0.001 (0.009) [0.927]	0.040** (0.013) [0.043]	-0.005 (0.006) [0.524]	0.001 (0.006) [0.922]
Mean of dep. var. (pre-reform)	13.841	13.351	12.505	13.021	7.734
Observations	336	336	336	312	312
<i>Panel B: Manufacturing only (SIC 15–36)</i>					
TVM	0.010 (0.010) [0.364]	-0.003 (0.010) [0.724]	0.040* (0.014) [0.069]	-0.009 (0.007) [0.317]	-0.005 (0.007) [0.548]
Mean of dep. var. (pre-reform)	13.051	12.402	12.080	13.107	7.836
Observations	336	336	336	312	312
<i>Panel C: Agriculture only (SIC 1, 2, 5)</i>					
TVM	-0.007 (0.013) [0.866]	0.041** (0.009) [0.018]	0.036* (0.027) [0.095]	-0.001 (0.009) [0.901]	-0.020 (0.010) [0.702]
Mean of dep. var. (pre-reform)	12.483	12.086	10.538	12.897	7.613
Observations	336	336	330	312	312
<i>Panel D: Mining only (SIC 10–14)</i>					
TVM	-0.030 (0.034) [0.586]	0.037 (0.109) [0.728]	-0.034 (0.047) [0.451]	-0.074 (0.022) [0.128]	-0.067 (0.026) [0.265]
Mean of dep. var. (pre-reform)	10.085	9.188	9.364	13.711	8.365
Observations	332	302	303	308	308
Department FE	Yes	Yes	Yes	Yes	Yes
Year-semester FE	Yes	Yes	Yes	Yes	Yes

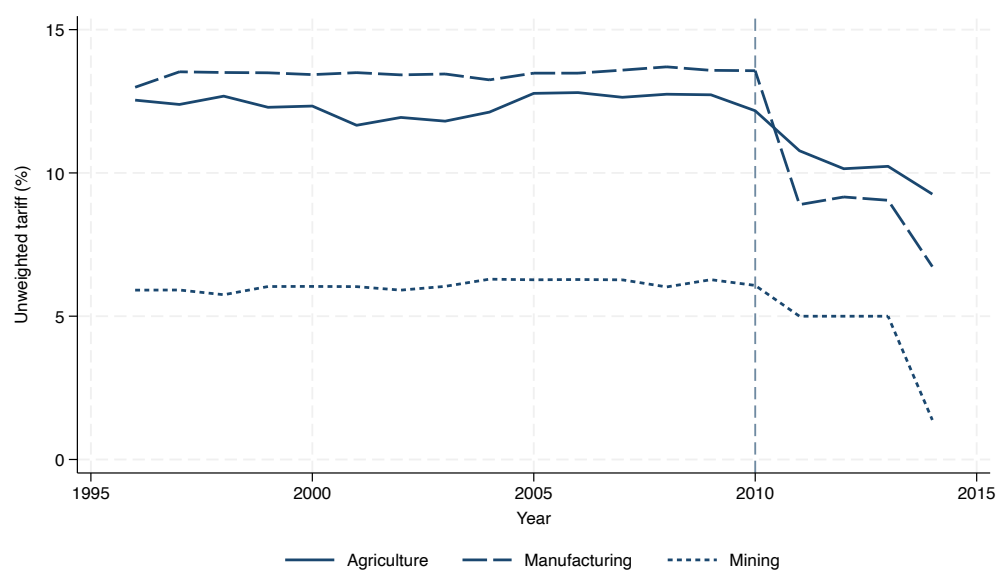
Note: Each column reports the coefficient on the time-varying tradable *TVM* from a separate regression with a labor outcome of prime-age workers (ages 25–44) as the dependent variable. Outcomes are constructed from the GEIH household survey aggregated to the department–semester level: column (1) is log total employment in the relevant sector; column (2) is log informal employment (workers without contributory pension coverage); column (3) is log formal employment; column (4) is the weighted mean log monthly wage among employed workers; column (5) is the weighted mean log hourly wage among employed workers. Panel A uses all tradable industries (agriculture, mining, and manufacturing); Panel B restricts to manufacturing; Panel C to agriculture; Panel D to mining. The treatment is the departmental Tariff Vulnerability Measure constructed from all-tradable employment shares, analogous to the municipal *TVM* in Table 1. The sample covers 2008s1–2014s2. All regressions include department and year-semester fixed effects, are weighted by 2009 working-age (18–65) population, and report standard errors clustered at the department level (in parentheses) and the wild cluster bootstrap *p*-value in square brackets, computed with Stata’s `wildbootstrap` command using 9,999 Rademacher-weight replications. Stars are assigned from the wild cluster bootstrap *p*-value: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figures

Figure 1: Evolution of Colombian tariffs, 1996–2014



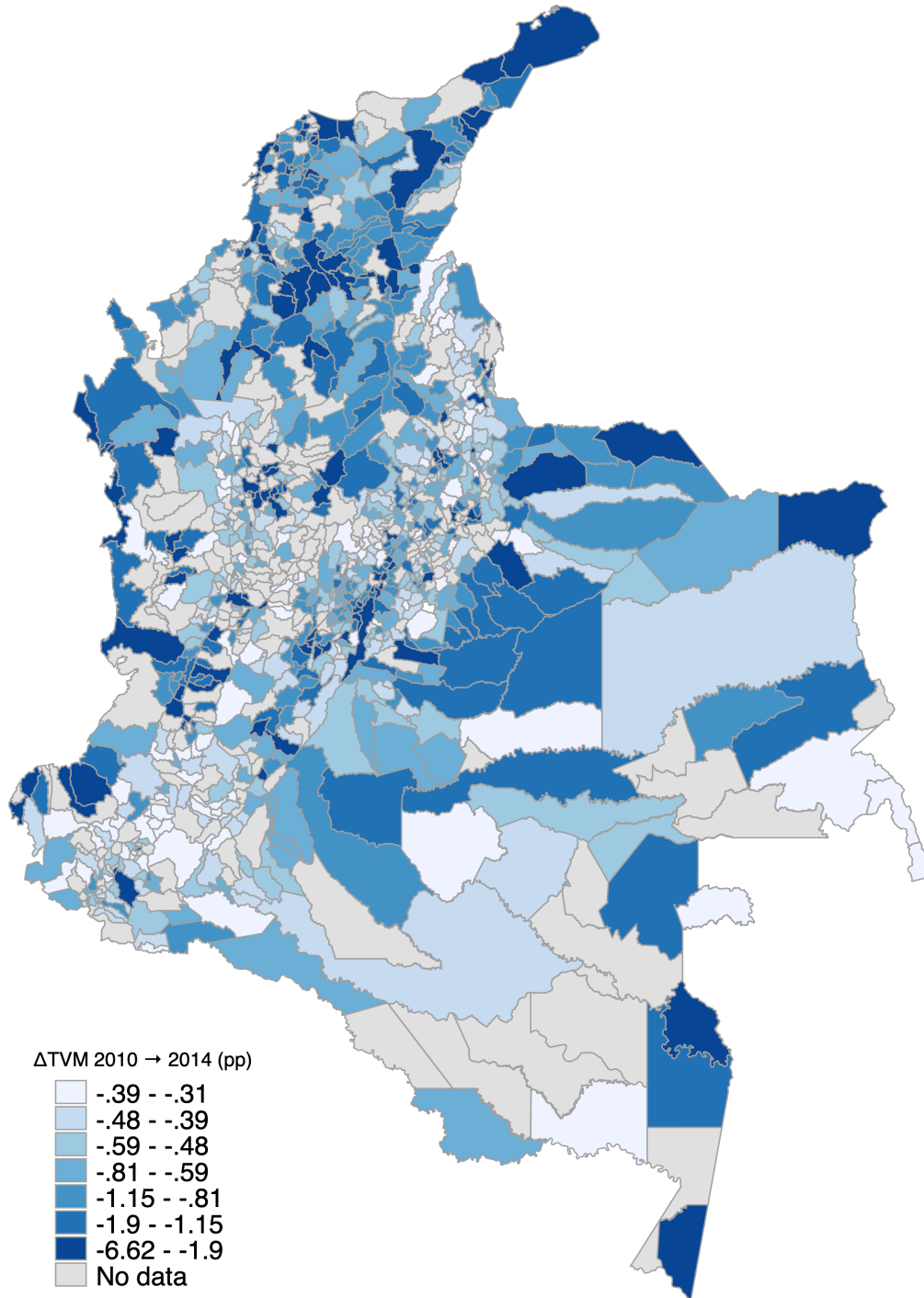
(a) All tradable sectors



(b) By economic sector

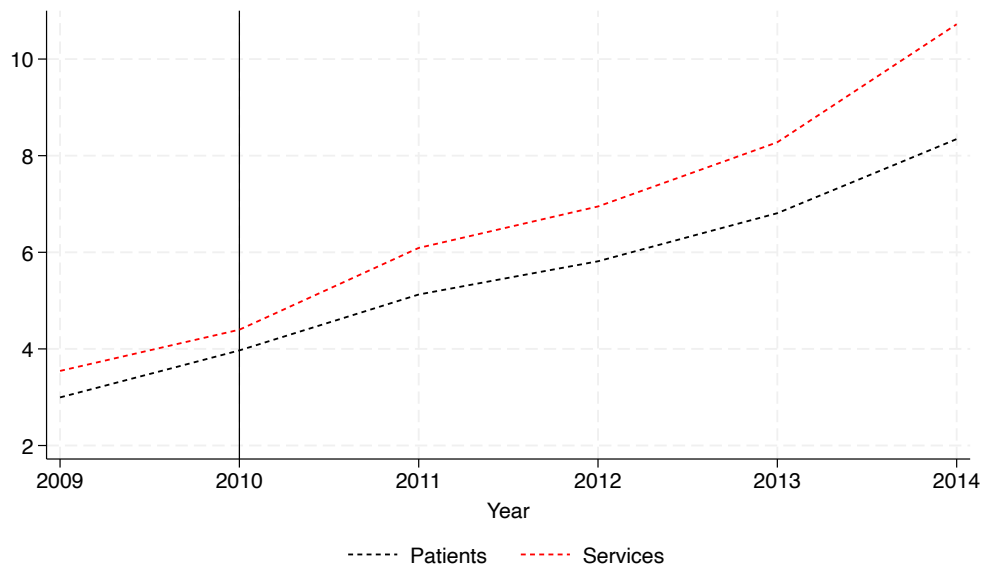
Note: Panel (a) plots the unweighted average ad-valorem tariff across all tradable industries by year. Panel (b) plots the unweighted average separately for agriculture, manufacturing, and mining. The vertical dashed line marks 2010, the year Decree 4114 was issued. Data source: WITS-WTO at the 2-digit SIC level.

Figure 2: Change in municipal TVM exposure from 2010 to 2014



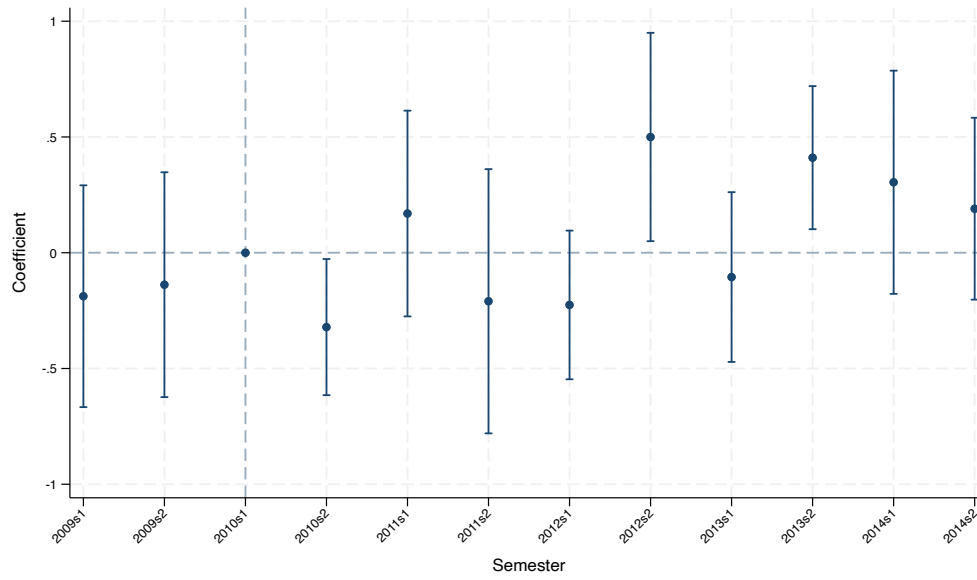
Note: The map shades each of the 842 municipalities in the substance-abuse estimation sample by the change in the Tariff Vulnerability Measure between 2010 and 2014, computed as $TVM_{m,2014} - TVM_{m,2010}$, in percentage points; negative values denote tariff-exposure reductions, and darker blue indicates a larger cut. The realized cut used elsewhere in the paper, $\Delta TVM_m = TVM_{m,2010} - TVM_{m,2014}$ (positive for cuts), is the additive inverse of the quantity plotted here. The treatment is the municipal TVM used in Table 1; year-level values average over semesters. Bin cutoffs are quantiles of the cross-municipality distribution.

Figure 3: Evolution of the substance-abuse rate, 2009–2014

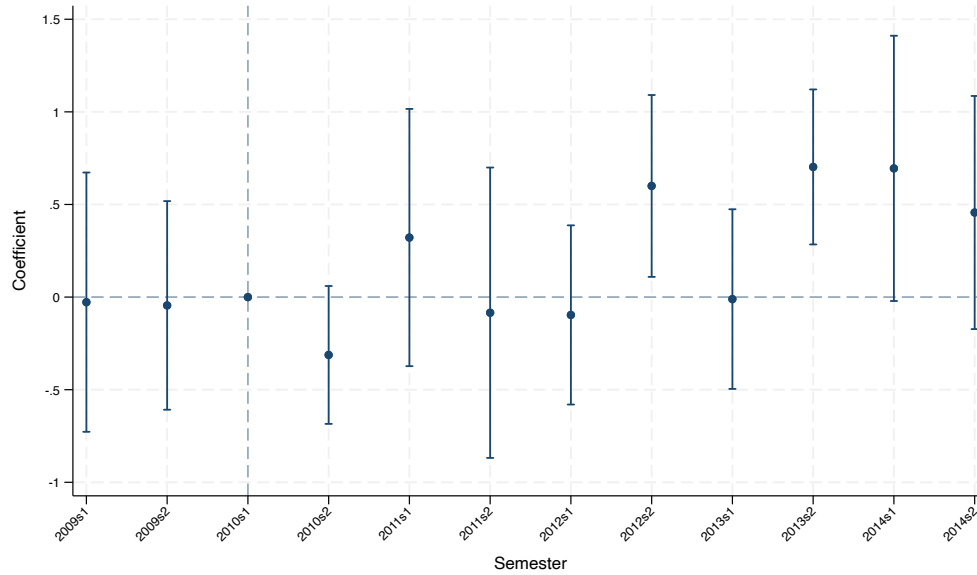


Note: The figure plots the annual population-weighted mean of the substance-abuse rate per 100,000 working-age population across the 842 municipalities in the estimation sample. The patient rate is the count of unique patients with an ICD-10 F10–F19 primary diagnosis; the service rate is the count of corresponding RIPS services delivered for those patients, covering hospitalizations and emergency-room visits (the two RIPS service categories used as outcomes throughout the paper). Annual values average over the two semesters. The vertical solid line marks 2010, the year of the unilateral tariff reform (Decree 4114).

Figure 4: Event-study estimates of the substance-abuse response to trade liberalization



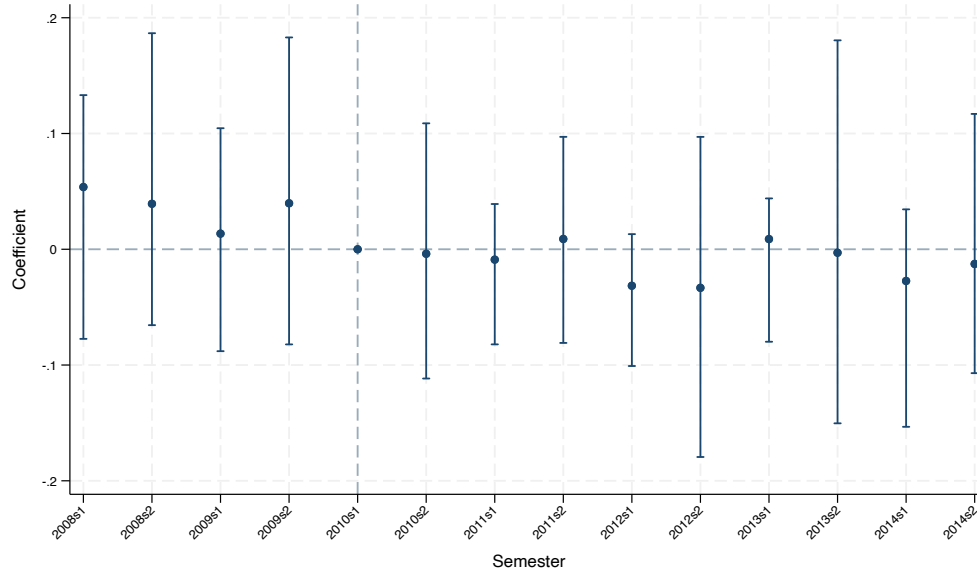
(a) Patient rate



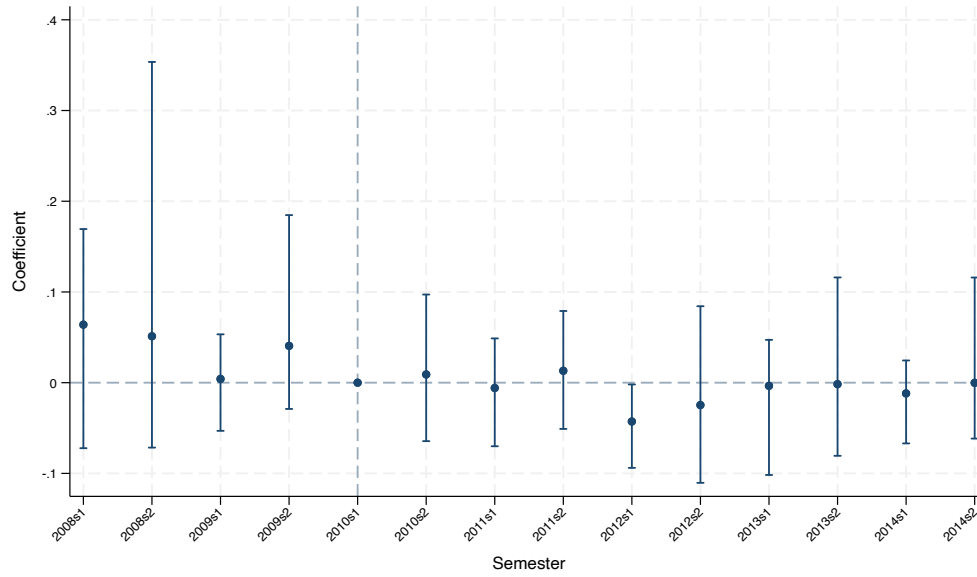
(b) Service rate

Note: The figure plots event-study coefficients from a regression of the substance-abuse case rate per 100,000 working-age population at the municipality–semester level on the standardized realized cut in the municipal Tariff Vulnerability Measure, $\widehat{\Delta TVM}_m$, interacted with year-semester indicators. The exposure measure is $\Delta TVM_m = TVM_{m,2010} - TVM_{m,2014}$, with positive values denoting larger tariff cuts. Panel (a) uses the patient rate; Panel (b) uses the service rate. The 2010s1 semester is the base period. Municipality and year-semester fixed effects are included, the post-LASSO union from the Belloni, Chernozhukov, and Hansen (2014) double-selection procedure is included as $\times Post$ controls, regressions are weighted by 2009 working-age (18–65) population, and standard errors are clustered at the municipality level. Whiskers show 95% confidence intervals. The dashed vertical line marks the base period; the dashed horizontal line marks zero.

Figure 5: Event-study estimates of the labor-market response to trade liberalization: log formal employment of prime-age (25–44) workers at the departmental level



(a) All tradable sectors



(b) Manufacturing only

Note: The figure plots event-study coefficients from a regression of log prime-age (ages 25–44) formal employment at the department–semester level on the standardized realized cut in the all-tradable departmental Tariff Vulnerability Measure, ΔTVM_d , interacted with year-semester indicators. The exposure measure is $\Delta TVM_d = TVM_{d,2010} - TVM_{d,2014}$, with positive values denoting larger tariff cuts. The 2010s1 semester is the base period. Department and year-semester fixed effects are included. Whiskers show wild cluster bootstrap 95% confidence intervals computed with Stata’s `wildbootstrap` command using 9,999 Rademacher-weight replications, clustering at the department level. Regressions are weighted by 2009 working-age population. The dashed vertical line marks the base period; the dashed horizontal line marks zero.

Online Appendix

A Supplementary Tables and Figures

Table A1: Distribution of substance-abuse outcomes and baseline municipal characteristics

	Mean (1)	SD (2)	% Zero (3)	Median (4)	Max (5)	N (6)
<i>Panel A.1: Patient rate per 100,000 (muni-semester)</i>						
Total substance abuse	2.36	8.15	83.3	0.00	164.90	10,104
Alcohol	1.13	6.45	91.6	0.00	164.90	10,104
Hallucinogens	0.26	2.05	96.1	0.00	51.12	10,104
Cannabinoids	0.07	1.38	98.5	0.00	101.73	10,104
Cocaine	0.07	0.86	98.2	0.00	32.10	10,104
Volatile solvents	0.18	1.79	97.4	0.00	55.68	10,104
Multiple drugs	0.38	2.49	94.8	0.00	95.42	10,104
Opiates	0.06	0.85	98.5	0.00	41.00	10,104
Other stimulants	0.05	0.78	98.6	0.00	31.90	10,104
Sedatives	0.17	1.54	96.6	0.00	63.37	10,104
<i>Panel A.2: Service rate per 100,000 (muni-semester)</i>						
Total substance abuse	2.87	10.61	83.3	0.00	190.84	10,104
Alcohol	1.34	8.21	91.6	0.00	177.05	10,104
Hallucinogens	0.34	2.95	96.1	0.00	100.17	10,104
Cannabinoids	0.08	1.44	98.5	0.00	101.73	10,104
Cocaine	0.08	0.95	98.2	0.00	32.10	10,104
Volatile solvents	0.21	2.24	97.4	0.00	91.98	10,104
Multiple drugs	0.47	3.61	94.8	0.00	190.84	10,104
Opiates	0.07	1.33	98.5	0.00	82.00	10,104
Other stimulants	0.06	1.00	98.6	0.00	48.56	10,104
Sedatives	0.20	2.12	96.6	0.00	126.74	10,104
<i>Panel B: Baseline municipal characteristics (time-invariant, muni level)</i>						
Log working-age population (2009)	8.83	1.07	–	8.76	15.31	842
Pre-period SA rate (2009s1)	1.14	4.86	89.7	0.00	56.24	842
Log RIPS service volume (2009s1)	4.41	1.48	–	4.32	11.87	833
Unsatisfied Basic Needs (NBI)	46.32	20.56	–	44.16	100.00	842
Pre-period homicide rate (per 100k)	767.15	1752.19	22.2	244.25	21322.87	842
Public investment share in health	0.38	0.14	0.4	0.39	0.77	829

Note: Panels A.1 and A.2 report distributional statistics for the substance-abuse outcomes across the 10,104 municipality–semester observations in the estimation sample, separately for the patient rate (unique patients per 100,000 working-age population) and the service rate (services delivered per 100,000). The first row of each panel is the total substance-abuse rate; the remaining rows correspond to the nine ICD-10 substance categories. The “% Zero” column reports the share of municipality–semester observations with no recorded cases. Panel B reports distributional statistics across the 842 municipalities for time-invariant baseline characteristics that constitute the candidate pool for the post-LASSO control selection in the main specification. Per-capita and per-100,000 measures use the 2009 working-age population as the denominator. The “% Zero” column is reported only where the variable can take exact zero values; “–” indicates a strictly continuous variable.

Table A2: Summary statistics: industry-level tariffs and municipal TVM exposure

	Mean (1)	SD (2)	Min (3)	P25 (4)	Median (5)	P75 (6)	Max (7)	N (8)
<i>Panel A: 2-digit SIC tariffs (unweighted)</i>								
Tariff in 2010 (%)	12.39	4.81	5.00	8.87	12.45	15.41	20.00	29
Tariff in 2014 (%)	6.25	4.62	0.00	2.41	5.31	9.75	16.36	29
Tariff change, 2010 → 2014 (pp)	-6.13	2.36	-11.44	-7.50	-6.12	-4.76	-0.31	29
<i>Panel B: Municipal TVM exposure (weighted by 2009 working-age population)</i>								
TVM in 2010 (%)	13.87	2.20	7.83	11.47	14.19	15.64	17.77	842
TVM in 2014 (%)	10.31	0.55	4.78	10.09	10.30	10.52	12.76	842
TVM change, 2010 → 2014 (pp)	-3.56	2.19	-6.62	-5.54	-4.80	-1.16	-0.31	842
<i>Panel C: Department TVM exposure (weighted by 2009 working-age population)</i>								
TVM in 2010 (%)	13.27	1.84	10.98	11.66	13.50	13.85	16.01	32
TVM in 2014 (%)	10.45	0.21	9.42	10.34	10.42	10.62	10.81	32
TVM change, 2010 → 2014 (pp)	-2.81	1.84	-5.68	-3.14	-2.69	-1.30	-0.56	32

Note: Panel A reports unweighted summary statistics across the 29 tradable 2-digit SIC rev. 3 industries for the WITS-WTO effectively applied tariff in 2010 and 2014 and for the corresponding change in percentage points. Panel B reports population-weighted summary statistics across the 842 municipalities in the substance-abuse estimation sample for the municipal Tariff Vulnerability Measure in 2010 and 2014 and for the corresponding change, using 2009 working-age (18–65) population as weights. Panel C reports the same statistics across the 32 departments used in the labor-mechanism analysis, where the department-level Tariff Vulnerability Measure is constructed from 2005 department-level employment shares and the same time-varying industry tariff. Annual values are computed as the within-year mean of the semestral observations.

Table A3: ICD classification code and substance name

ICD code	Substance name
F100–F101	Alcohol
F110–F111	Opiates
F120–F121	Cannabinoids
F130–F131	Sedatives or hypnotics
F140–F141	Cocaine
F150–F151	Other stimulants, including caffeine
F160–F161	Hallucinogens
F170–F171	Tobacco
F180–F181	Volatile solvents
F190–F191	Multiple drugs and use of other psychoactive substances

Note: Each of the substances listed belongs to the general category of the ICD “F10–F19. Mental and behavioural disorders due to use of psychoactive substances: acute intoxication and harmful use.” Total substance abuse is calculated by summing each category and scaling by working-age (18–65) population per 100,000. Tobacco (F170–F171) is excluded from the substance-by-substance regressions reported in Table 3 because of negligible within-municipality variation over the sample period.

Table A4: Robustness: added controls

	(1)	(2)	(3)	(4)
<i>Panel A: Patient rate (per 100,000)</i>				
TVM (tradable)	-0.424*** (0.130)	-0.369** (0.174)	-0.430*** (0.137)	-0.699*** (0.132)
<i>Panel B: Service rate (per 100,000)</i>				
TVM (tradable)	-0.622*** (0.181)	-0.607** (0.258)	-0.625*** (0.189)	-0.917*** (0.168)
Added control	–	Partner tariff	Input tariff	–
Mean of dep. var. (pre-reform)	3.93	3.93	3.93	3.93
Observations	9,948	9,948	9,948	9,948
Municipalities	829	829	829	829
Municipality FE	Yes	Yes	Yes	Yes
Year-semester FE	Yes	Yes	Yes	Yes
Department × year FE	No	No	No	Yes

Note: Each column re-estimates the post-double-LASSO main specification on the substance-abuse case rate per 100,000 working-age population at the municipality–semester level. Panel A uses the patient rate; Panel B uses the service rate. Column (1) reports the baseline. Column (2) adds the employment-weighted tariff on Colombia’s exports to its main trading partners as an additional regressor in levels. Column (3) adds the muni-year input tariff constructed following [Edmonds et al. \(2010\)](#). Column (4) adds department-by-year fixed effects to the baseline. Baseline controls interacted with a post-reform indicator are selected via the [Belloni et al. \(2014\)](#) post-double-LASSO procedure from a four-variable candidate pool: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age (18–65) population, and report standard errors clustered at the municipality level (in parentheses). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A5: Sensitivity of Table 1 to the BCH double-LASSO candidate pool

	(1)	(2)	(3)	(4)
	Patients		Services	
TVM (tradable)	-0.424*** (0.130)	-0.329** (0.151)	-0.622*** (0.181)	-0.496** (0.223)
Mean of dep. var. (pre-reform)	3.40	3.40	3.93	3.93
Observations	9,948	9,888	9,948	9,888
Municipalities	829	824	829	824
Candidate pool	4 cand	6 cand	4 cand	6 cand
Selected (BCH union)	4 / 4	5 / 6	4 / 4	5 / 6

Note: The table reports the *TVM* coefficient on the time-varying tradable *TVM* from the post-double-LASSO main specification under two alternative candidate pools for the [Belloni et al. \(2014\)](#) selection step. Columns (1) and (3) use the main four-variable candidate pool: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000, each interacted with $Post_t$. Columns (2) and (4) augment the pool with two variables that are mechanically correlated with baseline *TVM* under the shift-share construction: log RIPS service volume in 2009s1 (which scales with municipal health-system reporting capacity) and log 2009 working-age population (which scales with the industrial diversification underlying the Bartik exposure measure). All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age population, and report standard errors clustered at the municipality level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A6: Robustness: alternative samples

	(1)	(2)	(3)
<i>Panel A: Patient rate (per 100,000)</i>			
TVM (tradable)	-0.424*** (0.130)	-0.404*** (0.146)	-0.446*** (0.130)
<i>Panel B: Service rate (per 100,000)</i>			
TVM (tradable)	-0.622*** (0.181)	-0.599*** (0.198)	-0.649*** (0.181)
Sample	Full	No coca	Trimmed
Mean of dep. var. (pre-reform)	3.93	4.13	3.93
Observations	9,948	8,328	9,636
Municipalities	829	694	803
Municipality FE	Yes	Yes	Yes
Year-semester FE	Yes	Yes	Yes

Note: Each column re-estimates the post-double-LASSO main specification on the substance-abuse case rate per 100,000 working-age population at the municipality–semester level. Panel A uses the patient rate; Panel B uses the service rate. Column (1) reports the baseline on the full estimation sample of 842 municipalities. Column (2) restricts the sample to the 694 municipalities with no historical coca cultivation in 2005. Column (3) trims the sample to municipalities with values of the 2010–2014 tariff cut $\Delta TVM_m = TVM_{m,2010} - TVM_{m,2014}$ between the population-weighted first and ninety-ninth percentiles. In all three columns, baseline controls interacted with a post-reform indicator are selected via the Belloni et al. (2014) post-double-LASSO procedure from a four-variable candidate pool: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age (18–65) population, and report standard errors clustered at the municipality level (in parentheses). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A7: Robustness to the zero-inflated outcome distribution

	OLS rate / log(1+rate)			PPML IRR		Heterogeneity diagnostics		
	Rate Full (1)	log(1+rate) Full (2)	Rate Above-med-pop (3)	Counts Full (4)	Counts Above-med-pop (5)	OLS rate PPML smp. (6)	OLS rate Full -5 (7)	PPML IRR Full -5 (8)
<i>Panel A: Patients per 100,000 working-age population</i>								
TVM (tradable)	-0.525*** (0.081)	-0.047** (0.022)	-0.471*** (0.094)	1.307*** (0.071)	1.310*** (0.071)	-0.386*** (0.097)	-0.489* (0.250)	1.146 (0.099)
<i>Panel B: Services per 100,000 working-age population</i>								
TVM (tradable)	-0.715*** (0.108)	-0.057** (0.024)	-0.638*** (0.123)	1.365*** (0.090)	1.370*** (0.090)	-0.530*** (0.125)	-0.664** (0.331)	1.146 (0.110)
Observations	10,104	10,104	5,052	6,120	4,212	6,120	10,044	6,060
Municipalities	842	842	421	510	351	510	837	505

Note: Each cell reports the coefficient on the time-varying tradable *TVM* from a separate regression. Column (1) reports the OLS estimate on the per-100,000 case rate on the full estimation sample. Column (2) replaces the rate with $\log(1 + \text{rate})$. Column (3) restricts the sample to the municipalities with above-median 2009 working-age population. Columns (4) and (5) report a pseudo-maximum-likelihood Poisson (PPML) estimator on the case count with $\log(\text{pop})$ entered as an offset (Silva and Tenreiro, 2006), with the coefficient displayed as an incidence-rate-ratio; column (4) uses the full sample and column (5) restricts to the above-median-population sample. The PPML specification includes municipality fixed effects, which drops municipalities whose count is zero in every semester. Column (6) re-estimates the OLS rate specification on the same set of municipalities retained by the PPML estimator in column (4). Column (7) re-estimates the OLS rate specification, and column (8) re-estimates the PPML specification, after dropping the five municipalities with the largest total recorded substance-abuse case counts over 2009–2014: Bogotá, Medellín, Cali, Bucaramanga, and Cartagena. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age (18–65) population, and report standard errors clustered at the municipality level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A8: Sensitivity of the *TVM* coefficient to dropping the highest-weighted industries in the Goldsmith-Pinkham, Sorkin, and Swift (2020) Rotemberg decomposition

Industries dropped	TVM coef. (patients) (1)	TVM coef. (services) (2)	N (3)
(baseline)	-0.424*** (0.130)	-0.622*** (0.181)	9,948
18	-0.497*** (0.159)	-0.812*** (0.224)	9,948
18, 17	-0.662*** (0.203)	-1.176*** (0.282)	9,948
18, 17, 15	-0.690*** (0.238)	-1.280*** (0.335)	9,948
18, 17, 15, 28	-0.738*** (0.272)	-1.375*** (0.402)	9,948
18, 17, 15, 28, 22	-0.772*** (0.299)	-1.467*** (0.441)	9,948

Note: Each row re-estimates the post-double-LASSO main specification on the substance-abuse case rate per 100,000 working-age population at the municipality–semester level, with the treatment $TVM_{m,y(t)}$ rebuilt after dropping the listed industries from its construction. The first row reports the baseline using all 29 tradable industries; each subsequent row drops the next industry in order of Rotemberg weight from the Goldsmith-Pinkham et al. (2020) decomposition. The industries dropped are SIC rev. 3 codes 18 (wearing apparel), 17 (textiles), 15 (food products and beverages), 28 (fabricated metal products), and 22 (publishing and printing). Column (1) reports the *TVM* coefficient on the patient rate; column (2) reports the *TVM* coefficient on the service rate; column (3) reports the sample size. Baseline controls interacted with a post-reform indicator are selected via the Belloni et al. (2014) post-double-LASSO procedure from a four-variable candidate pool: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. The selected union from the baseline specification is held fixed across the six rows. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age (18–65) population, and report standard errors clustered at the municipality level (in parentheses). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A9: Robustness: most-favored-nation tariffs in place of effectively-applied tariffs

	(1)	(2)	(3)	(4)
	Patients		Services	
TVM (AHS)	-0.424***		-0.622***	
	(0.130)		(0.181)	
TVM (MFN)		-0.396***		-0.583***
		(0.130)		(0.195)
Mean of dep. var. (pre-reform)	3.40	3.40	3.93	3.93
Observations	9,948	9,948	9,948	9,948
Municipalities	829	829	829	829
Municipality FE	Yes	Yes	Yes	Yes
Year-semester FE	Yes	Yes	Yes	Yes
Tariff measure	AHS	MFN	AHS	MFN

Note: The table re-estimates the post-double-LASSO main specification with the TVM measure reconstructed from the most-favored-nation (MFN) applied tariff in place of the effectively-applied (AHS) tariff. The MFN series excludes the preferential tariff reductions granted under the Colombia–United States free trade agreement (effective May 2012) and the Colombia–European Union free trade agreement (effective August 2013), so an MFN-based TVM reflects the 2010 unilateral reform and other non-preferential variation in industry tariffs. Columns (1) and (3) reproduce the AHS-based estimate on the matched sample for comparison; columns (2) and (4) report the MFN-based estimate. The dependent variable is the per-100,000 working-age (18–65) rate of substance-abuse patients in columns (1)–(2) and substance-abuse services in columns (3)–(4). Baseline controls interacted with a post-reform indicator are selected via the [Belloni et al. \(2014\)](#) post-double-LASSO procedure from a four-variable candidate pool: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age population, and report standard errors clustered at the municipality level (in parentheses). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A10: Placebo: acute conditions sharing the RIPS reporting channel

	Infectious diseases (1)	Circulatory diseases (2)	Digestive diseases (3)	Congenital malformations (4)	External causes (5)
<i>Panel A: Patient rate (per 100,000)</i>					
TVM (tradable)	-10.729 (10.939)	-1.826** (0.800)	-2.718 (4.808)	0.058 (0.200)	0.295* (0.155)
<i>Panel B: Service rate (per 100,000)</i>					
TVM (tradable)	-8.936 (12.287)	-2.424** (1.101)	-0.343 (6.442)	0.030 (0.294)	0.372* (0.199)
Mean of dep. var. (pre-reform)	217.41	24.76	145.82	5.36	3.82
Observations	9,471	9,471	9,471	9,471	9,471
Municipalities	828	828	828	828	828
Municipality FE	Yes	Yes	Yes	Yes	Yes
Year-semester FE	Yes	Yes	Yes	Yes	Yes

Note: Each column re-estimates the post-double-LASSO main specification with the indicated placebo outcome as the dependent variable, expressed as the per-100,000 working-age (18–65) population rate using the same 2009 population denominator as the substance-abuse outcome. The placebo outcomes are five ICD-10 chapters available in RIPS at the municipality–semester level: certain infectious and parasitic diseases (A00–B99, Column 1), diseases of the circulatory system (I00–I99, Column 2), diseases of the digestive system (K00–K93, Column 3), congenital malformations, deformations and chromosomal abnormalities (Q00–Q99, Column 4), and external causes of morbidity and mortality including accidents and injuries (V01–Y98, Column 5). Panel A uses the patient rate; Panel B uses the service rate. Baseline controls interacted with a post-reform indicator are selected via the [Belloni et al. \(2014\)](#) post-double-LASSO procedure from a four-variable candidate pool: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age population, and report standard errors clustered at the municipality level (in parentheses). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A11: Substance-abuse response, by health-insurance regime

	Contributory (1)	Subsidized (2)
<i>Panel A: Patients per 100,000 working-age population</i>		
TVM (tradable)	-0.359*** (0.106)	-0.070 (0.103)
<i>Panel B: Services per 100,000 working-age population</i>		
TVM (tradable)	-0.537*** (0.164)	-0.095 (0.147)
Observations	9,948	9,948
Municipalities	829	829

Note: The dependent variable is the rate per 100,000 working-age (18–65) population at the municipality–semester level of substance-abuse patients (Panel A) or services (Panel B) with an ICD-10 F10–F19 primary diagnosis, restricted to the indicated health-insurance regime. The contributory regime enrolls formal employees and self-employed contributors; the subsidized regime covers people in conditions of vulnerability outside formal employment. The treatment is the time-varying Tariff Vulnerability Measure at the municipality level. All regressions include municipality and year-semester fixed effects, are weighted by 2009 working-age population, and report standard errors clustered at the municipality level (in parentheses). Baseline controls interacted with a post-reform indicator are included: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A12: Diagnostic: contributory share of substance-abuse patients

	Patients (1)	Services (2)
TVM (tradable)	-0.022* (0.012)	-0.023* (0.013)
Pre-reform mean of dep. var.	0.474	0.475
Observations	1,507	1,507
Municipalities	336	336

Note: The dependent variable is the contributory share within the substance-abuse patient cohort, $pat_{contr}/(pat_{contr} + pat_{subs})$ for column (1) and $serv_{contr}/(serv_{contr} + serv_{subs})$ for column (2), at the municipality–semester level. The sample is restricted to municipality–semesters with at least one F10–F19 case, since the share is otherwise undefined. The treatment is the time-varying Tariff Vulnerability Measure at the municipality level, in protection-level orientation. The specification matches the body regressions: municipality and year-semester fixed effects, the four-variable BCH control pool (pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000) interacted with a post-reform indicator, weighted by 2009 working-age (18–65) population, with standard errors clustered at the municipality level (in parentheses). A negative coefficient says the contributory share rises with a tariff cut, i.e. in the opposite direction to what a displacement-into-subsidized re-enrollment story would predict. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A13: Trade exposure and local coca cultivation

	Coca area (hectares) (1)	Coca plots (2)	Aerial eradication (3)	Manual eradication (4)	Total coca (5)
<i>Panel A: No controls</i>					
TVM (tradable)	0.362*** (0.106)	0.273*** (0.085)	0.091 (0.151)	-0.366 (0.280)	0.036 (0.040)
<i>Panel B: With four baseline controls $\times$ post</i>					
TVM (tradable)	0.408*** (0.098)	0.308*** (0.083)	0.182 (0.137)	-0.378 (0.286)	0.036 (0.045)
Observations	1,098	1,110	762	1,602	1,098
Municipalities	183	185	127	267	183
Municipality FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Controls $\times$ post	Yes	Yes	Yes	Yes	Yes

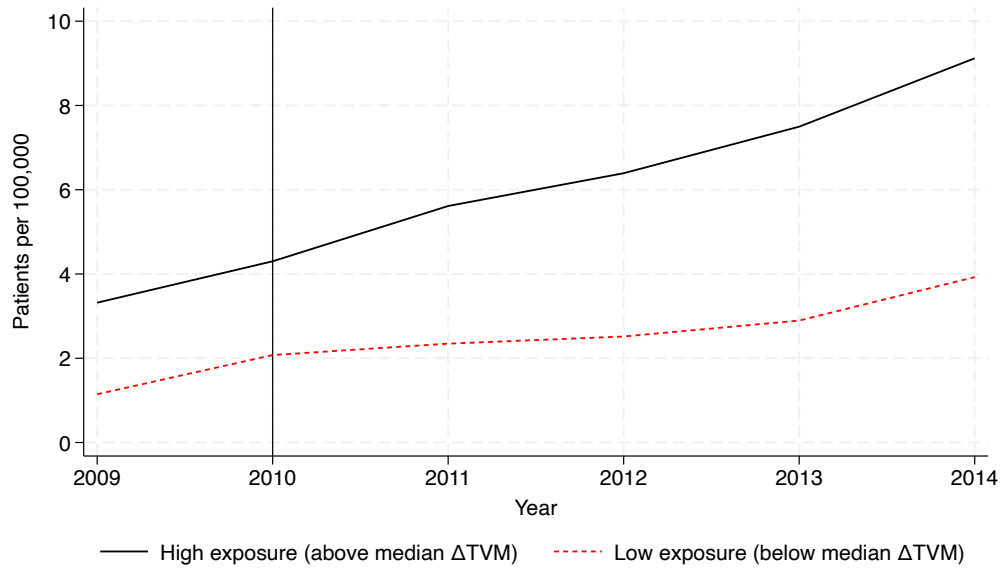
Note: Each column reports the Poisson coefficient on the time-varying Tariff Vulnerability Measure for one coca-cultivation outcome at the municipality-year level, 2009–2014. The outcomes come from the CEDE municipal conflict-and-violence panel and are: hectares of coca cultivated (Column 1), number of coca plots (Column 2), hectares of aerial coca eradication (Column 3), hectares of manual coca eradication (Column 4), and total hectares of coca cultivation (Column 5). Panel A omits baseline controls; Panel B adds four baseline characteristics interacted with a post-reform indicator: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. All regressions are unweighted Poisson estimated by [Correia et al. \(2020\)](#) with municipality and year fixed effects, and report standard errors clustered at the municipality level (in parentheses). Coefficients are semi-elasticities of the conditional mean of the outcome with respect to the treatment. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A14: Trade exposure and working-age population

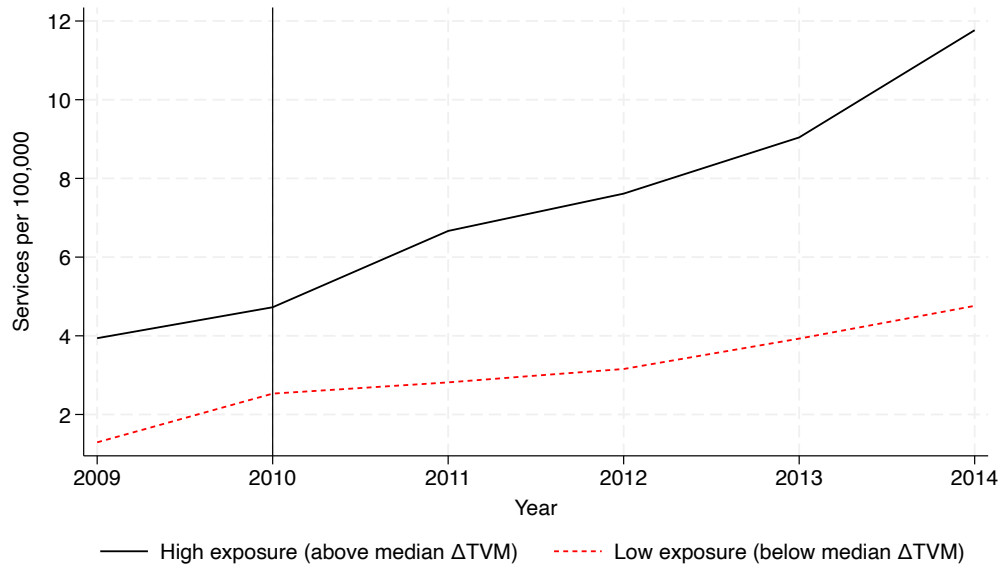
	(1)	(2)
TVM (tradable)	-0.0004 (0.0014)	0.0028 (0.0023)
Observations	5,052	4,974
Municipalities	842	829
Municipality FE	Yes	Yes
Year FE	Yes	Yes
Controls $\times$ post	No	Yes

Note: The dependent variable is the log of the working-age (18–65) total population of the municipality, taken from the DANE municipal population projections (2005–2017). The treatment is the time-varying Tariff Vulnerability Measure at the municipality level. Column (1) reports the baseline; column (2) adds four baseline characteristics interacted with a post-reform indicator: pre-reform substance-abuse rate, unmet basic needs index, share of public investment in health, and homicide rate per 100,000. Both regressions include municipality and year fixed effects, are weighted by 2009 working-age population, and report standard errors clustered at the municipality level (in parentheses). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure A1: Substance-abuse rate by tariff-exposure group, 2009–2014



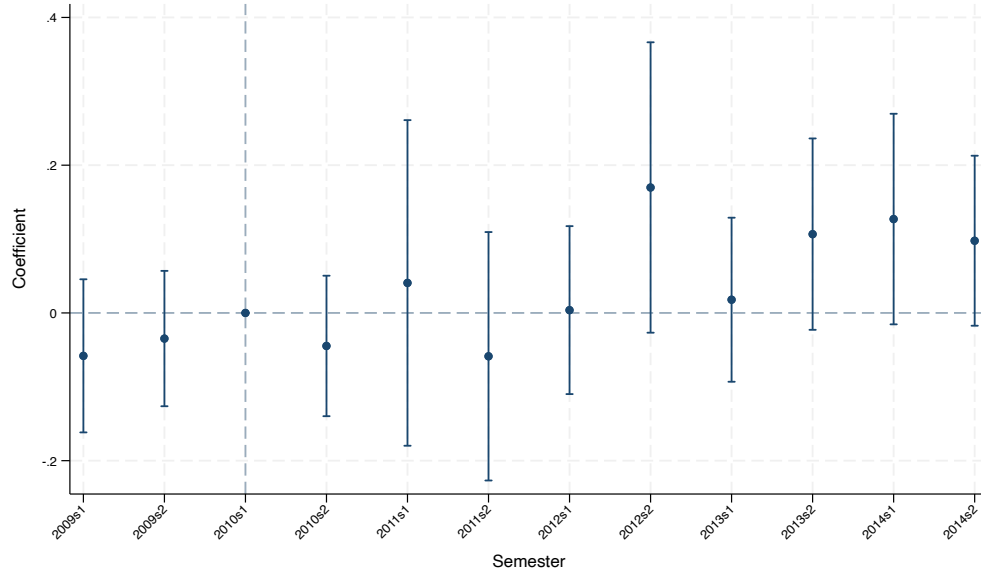
(a) Patient rate



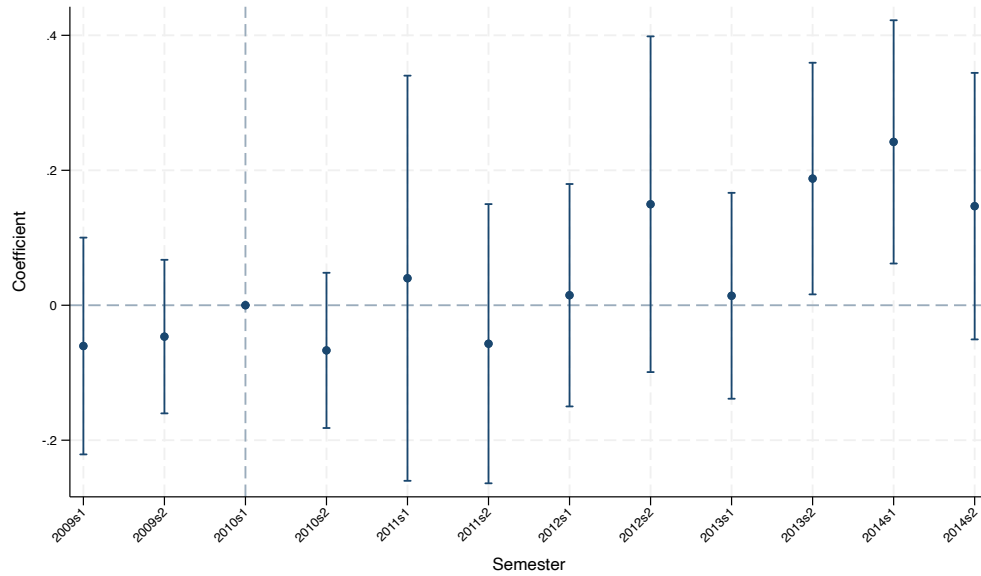
(b) Service rate

Note: The figure plots the annual population-weighted mean of the substance-abuse rate per 100,000 working-age population, separately for municipalities above the median realized tariff cut $\Delta TVM_m = TVM_{m,2010} - TVM_{m,2014}$ (high exposure, 421 munis) and below the median (low exposure, 421 munis). Panel (a) reports the patient rate; panel (b) reports the service rate. Annual values average over the two semesters. The vertical solid line marks 2010, the year of the unilateral tariff reform (Decree 4114).

Figure A2: Event-study estimates of the multiple-drug case rate



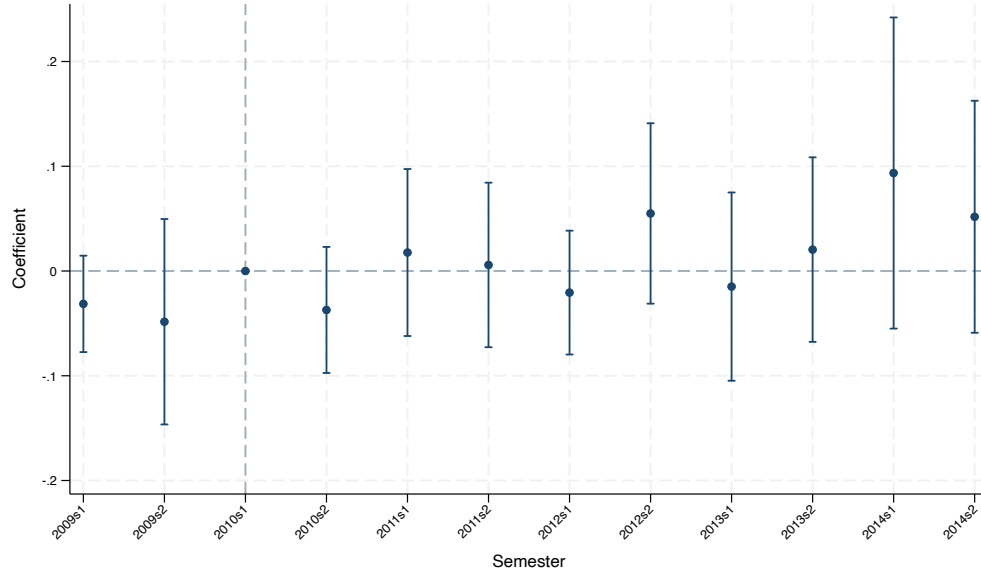
(a) Patient rate



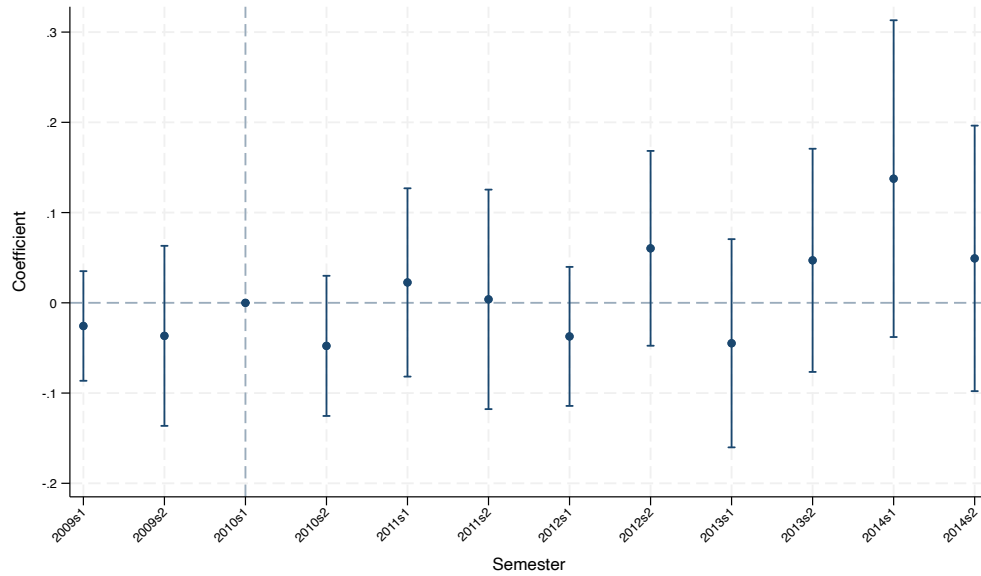
(b) Service rate

Note: The figure plots event-study coefficients from a regression of the per-100,000 case rate of multiple-drug use at the municipality–semester level on the standardized realized cut in the municipal Tariff Vulnerability Measure, $\widehat{\Delta TVM}_m$, interacted with year-semester indicators. The exposure measure is $\Delta TVM_m = TVM_{m,2010} - TVM_{m,2014}$, with positive values denoting larger tariff cuts. Panel (a) uses the patient rate; panel (b) uses the service rate. The 2010s1 semester is the base period. Municipality and year-semester fixed effects are included, the post-LASSO union from the Belloni, Chernozhukov, and Hansen (2014) double-selection procedure is included as $\times Post$ controls, regressions are weighted by 2009 working-age (18–65) population, and standard errors are clustered at the municipality level. Whiskers show 95% confidence intervals. The dashed vertical line marks the base period; the dashed horizontal line marks zero.

Figure A3: Event-study estimates of the sedative case rate



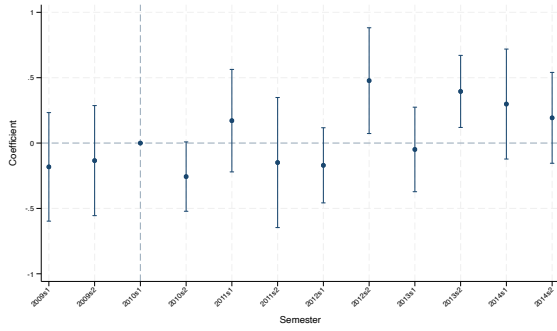
(a) Patient rate



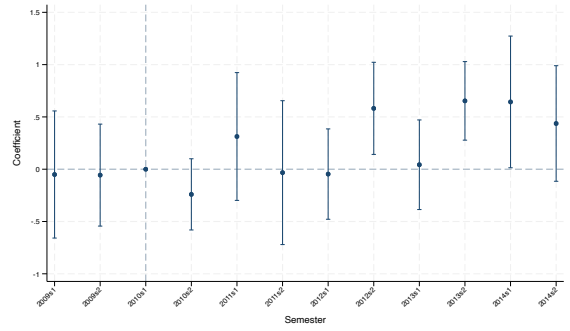
(b) Service rate

Note: The figure plots event-study coefficients from a regression of the per-100,000 case rate of sedative use at the municipality–semester level on the standardized realized cut in the municipal Tariff Vulnerability Measure, ΔTVM_m , interacted with year-semester indicators. The exposure measure is $\Delta TVM_m = TVM_{m,2010} - TVM_{m,2014}$, with positive values denoting larger tariff cuts. Panel (a) uses the patient rate; panel (b) uses the service rate. The 2010s1 semester is the base period. Municipality and year-semester fixed effects are included, the post-LASSO union from the Belloni, Chernozhukov, and Hansen (2014) double-selection procedure is included as $\times Post$ controls, regressions are weighted by 2009 working-age (18–65) population, and standard errors are clustered at the municipality level. Whiskers show 95% confidence intervals. The dashed vertical line marks the base period; the dashed horizontal line marks zero.

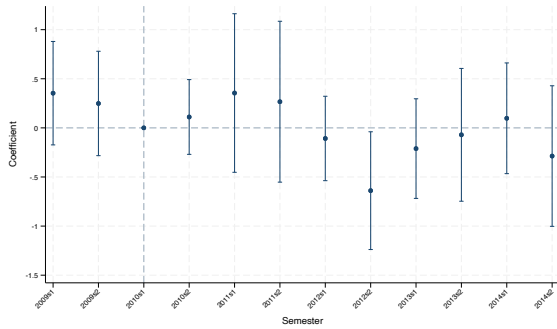
Figure A4: Event-study estimates of the substance-abuse response to sector-specific tariff cuts



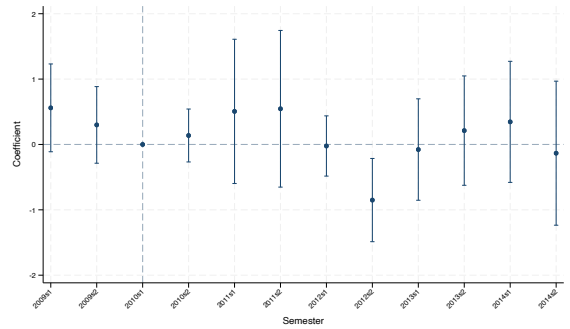
(a) Manufacturing, patient rate



(b) Manufacturing, service rate



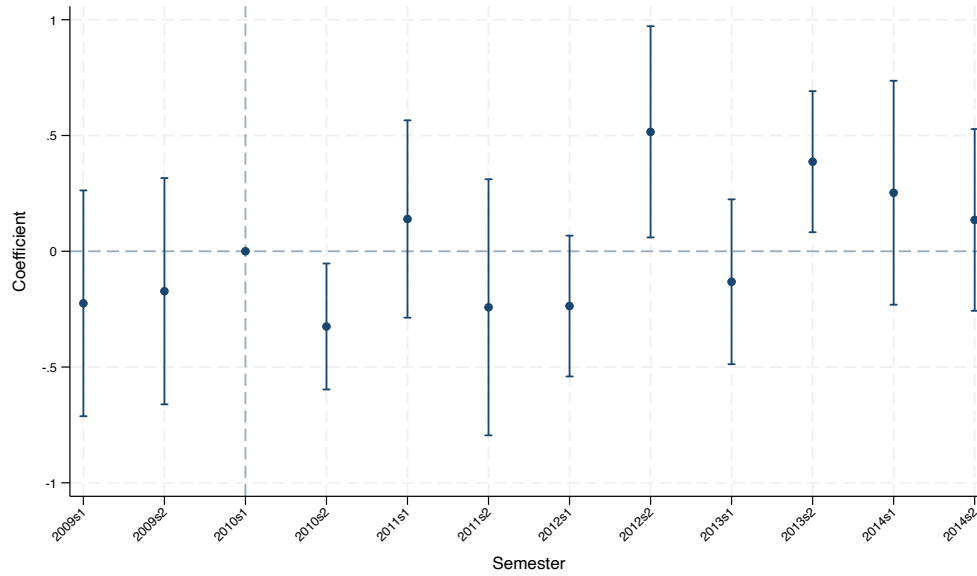
(c) Mining, patient rate



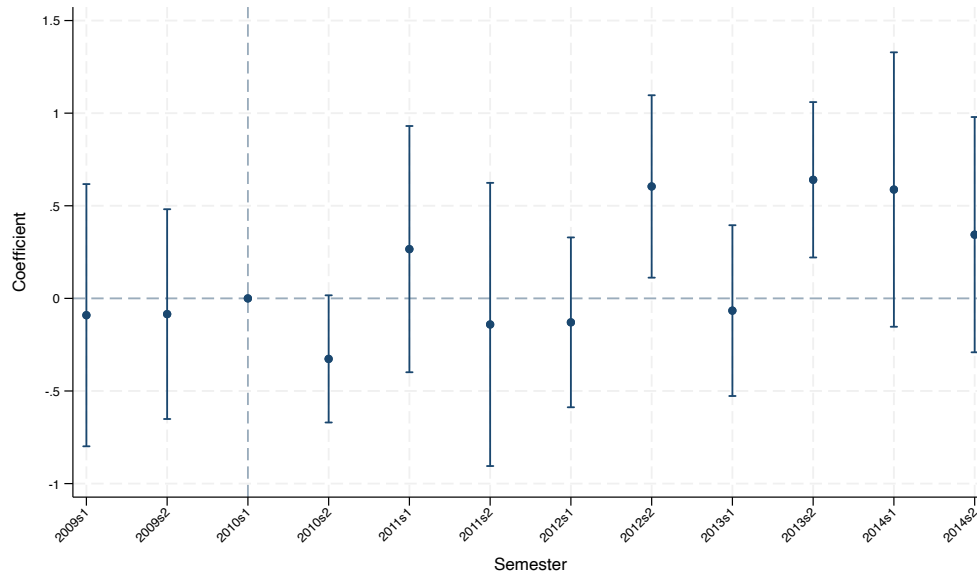
(d) Mining, service rate

Note: The figure plots event-study coefficients from regressions of the per-100,000 substance-abuse rate at the municipality–semester level on the standardized realized 2010–2014 sector-specific tariff cut, interacted with year-semester indicators. The sector-specific cut is constructed from 2005 employment shares in the sector’s two-digit SIC industries and the change in industry tariffs between 2010 and 2014; positive values denote larger tariff reductions. Panels (a)–(b) construct the cut from manufacturing industries only; panels (c)–(d) construct the cut from mining industries only. The 2010s1 semester is the base period. Municipality and year-semester fixed effects are included, the post-LASSO union from the Belloni, Chernozhukov, and Hansen (2014) double-selection procedure is included as $\times Post$ controls, regressions are weighted by 2009 working-age (18–65) population, and standard errors are clustered at the municipality level. Whiskers show 95% asymptotic cluster-robust confidence intervals. The dashed vertical line marks the base period; the dashed horizontal line marks zero.

Figure A5: Event-study estimates of the substance-abuse response to trade liberalization, MFN-based exposure



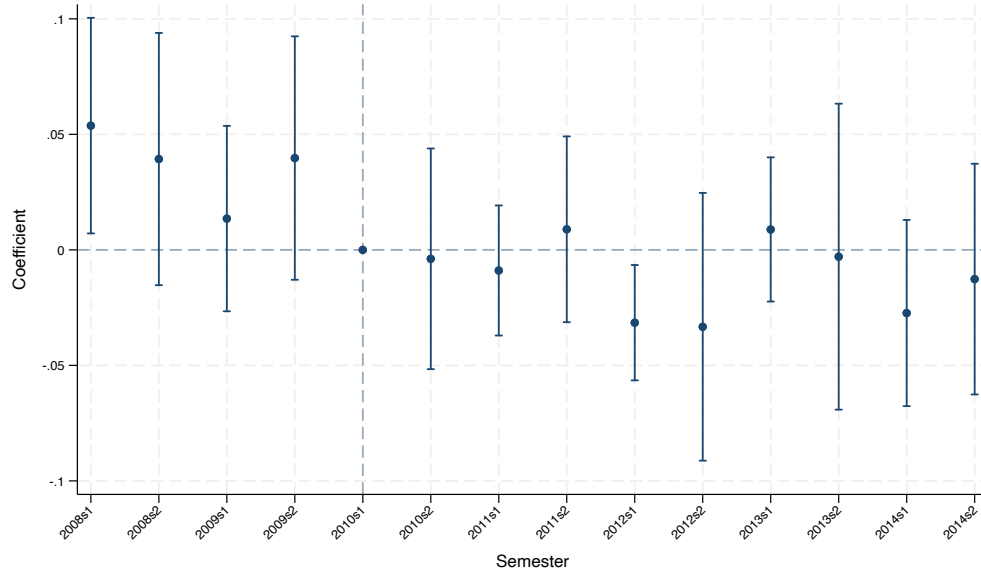
(a) Patient rate



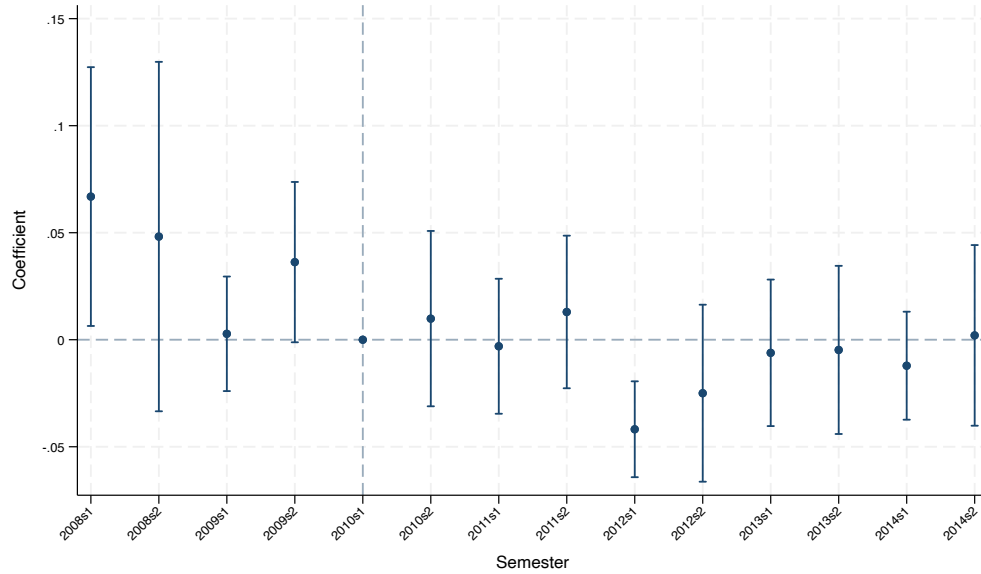
(b) Service rate

Note: The figure plots event-study coefficients from a regression of the substance-abuse case rate per 100,000 working-age population at the municipality–semester level on the standardized realized cut in the municipality Tariff Vulnerability Measure reconstructed from most-favored-nation (MFN) applied tariffs, interacted with year-semester indicators. The exposure measure is $\Delta TVM_m^{MFN} = TVM_{m,2010}^{MFN} - TVM_{m,2014}^{MFN}$, with positive values denoting larger tariff cuts. Panel (a) uses the patient rate; Panel (b) uses the service rate. The 2010s1 semester is the base period. Municipality and year-semester fixed effects are included, the same baseline controls used in the event study reported in the body are included as $\times$ Post, regressions are weighted by 2009 working-age (18–65) population, and standard errors are clustered at the municipality level. Whiskers show 95% confidence intervals. The dashed vertical line marks the base period; the dashed horizontal line marks zero.

Figure A6: Event-study estimates of the labor-market response to trade liberalization with asymptotic cluster-robust confidence intervals (companion to Figure 5)



(a) All tradable sectors



(b) Manufacturing only

Note: The figure plots event-study coefficients from a regression of log prime-age (ages 25–44) formal employment at the department–semester level on the standardized realized cut in the departmental Tariff Vulnerability Measure, ΔTVM_d , interacted with year-semester indicators. The exposure measure is $\Delta TVM_d = TVM_{d,2010} - TVM_{d,2014}$, with positive values denoting larger tariff cuts. Panel (a) uses the all-tradable departmental ΔTVM ; Panel (b) uses the manufacturing-only departmental ΔTVM . The 2010s1 semester is the base period. Department and year-semester fixed effects are included. Regressions are weighted by 2009 working-age population, and standard errors are clustered at the department level. Whiskers show asymptotic cluster-robust 95% confidence intervals. The dashed vertical line marks the base period; the dashed horizontal line marks zero.

B Validity of shift-share design

I test the GPSS share-exogeneity assumption directly by asking whether the 2005 industry employment shares predict the pre-reform dynamics of the substance-abuse outcome. Following Goldsmith-Pinkham et al. (2020), I first calculate the Rotemberg weights (Rotemberg, 1983), and focus on the five two-digit industries that carry the largest Rotemberg weights in the shift-share decomposition of $TVM_{m,y(t)}$: SIC 18 (wearing apparel), 17 (textiles), 15 (food products and beverages), 28 (fabricated metal products), and 22 (publishing and printing).⁴⁰ For each industry, I estimate the following regression by OLS:

$$\Delta Y_m = \alpha_j + \psi_j \tilde{s}_{m,j} + \varepsilon_{m,j},$$

where $\Delta Y_m = Y_{m,2010s1} - Y_{m,2009s1}$ is the pre-reform change in the substance-abuse rate of municipality m , $s_{m,j}$ is the 2005 employment share of industry j in municipality m , and $\tilde{s}_{m,j}$ standardizes the share to mean zero and unit variance using population-weighted moments across the estimation sample, matching the standardization of $\tilde{c}_m$ in the balance test below so that the two diagnostics are on a common scale. Regressions are weighted by 2009 working-age population and standard errors are heteroskedasticity-robust.

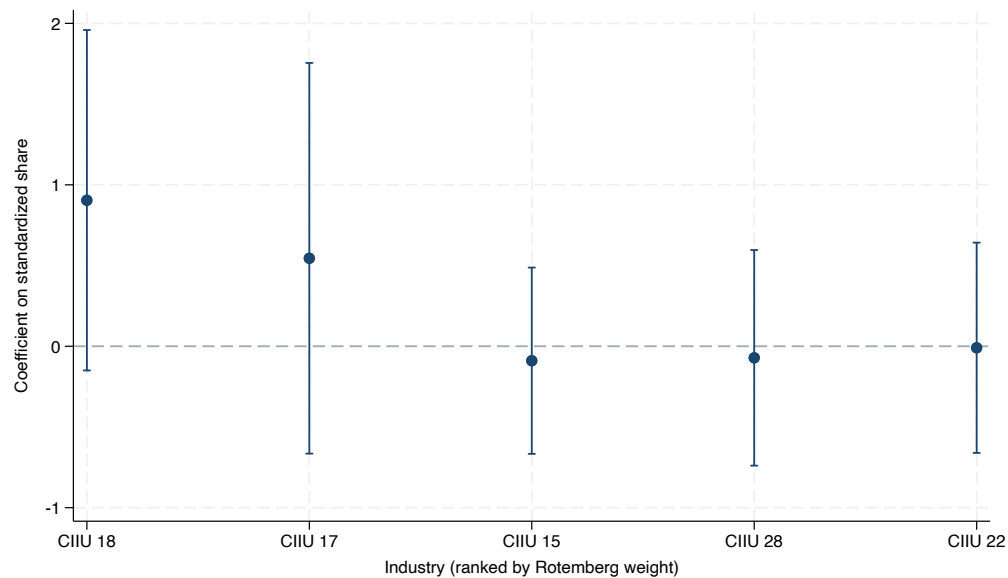
While the share-level test in Figure B1 supports the GPSS share-exogeneity assumption, I also examine whether the aggregated baseline exposure measure $TVM_{m,2010}$ is correlated with pre-reform municipality characteristics that independently predict substance-abuse outcomes. For each characteristic c in a set $\mathcal{C}$ of six baseline observables, I estimate by OLS:

$$\tilde{c}_m = \alpha_c + \theta_c \widetilde{TVM}_m + \eta_{m,c}, \quad c \in \mathcal{C}$$

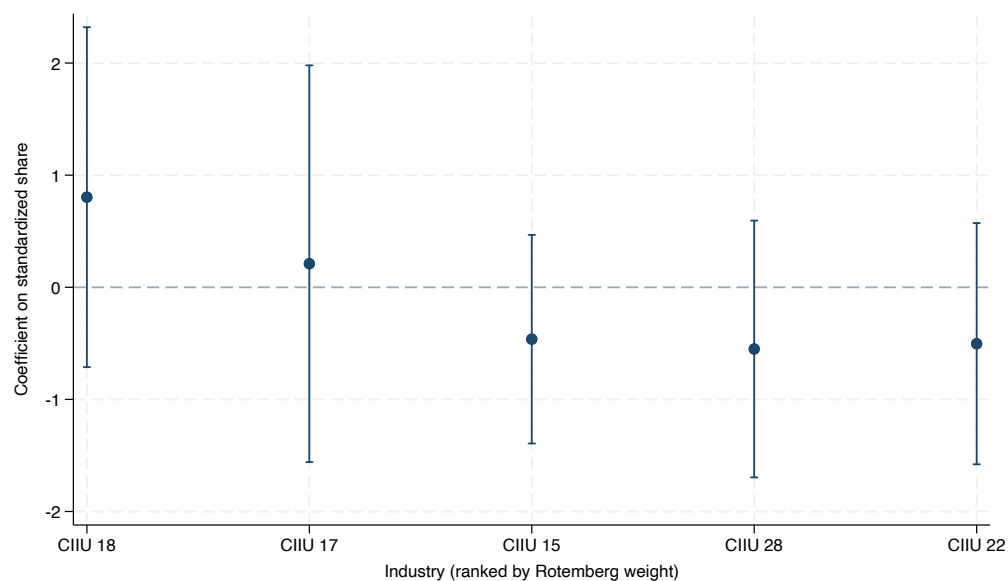
where $\widetilde{TVM}_m$ is the standardized 2010 value of $TVM_{m,y(t)}$ and $\tilde{c}_m$ standardizes

⁴⁰These five industries jointly account for approximately 69% of the positive Rotemberg weight in the decomposition.

Figure B1: Pre-trends test of the substance-abuse rate on the five highest-Rotemberg-weighted industry shares



(a) Patient rate



(b) Service rate

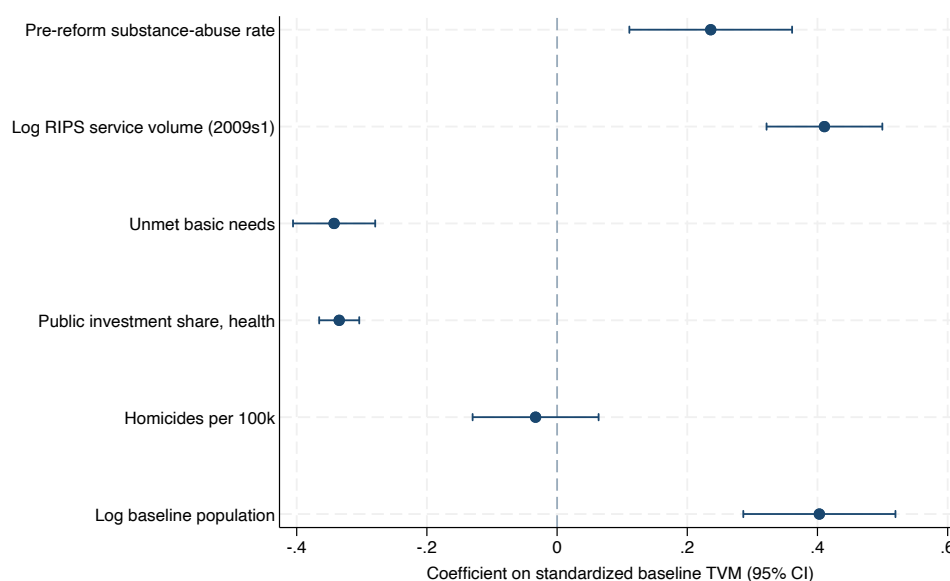
Note: Each marker reports the coefficient from a separate regression of the pre-reform change in the substance-abuse rate (the rate at 2010s1 minus the rate at 2009s1) on the standardized 2005 employment share of the indicated 2-digit SIC industry, with whiskers showing 95% confidence intervals from heteroskedasticity-robust standard errors. Regressions are weighted by 2009 working-age (18–65) population. Industries are ordered along the horizontal axis by their Rotemberg weight from the Goldsmith-Pinkham, Sorkin, and Swift (2020) decomposition; left to right these are SIC 18 (wearing apparel), 17 (textiles), 15 (food products and beverages), 28 (fabricated metal products), and 22 (publishing and printing). The dashed horizontal line marks zero.

the characteristic to mean zero and unit variance using population-weighted moments across the 824 municipalities in the estimation sample. The set C contains: the pre-reform substance-abuse rate, log RIPS service volume in 2009s1, the unmet basic needs index, the share of public investment in health, the homicide rate per 100,000, and log 2009 working-age population. Regressions are weighted by 2009 working-age population and standard errors are heteroskedasticity-robust.

Figure B2 reports the coefficients $\hat{\theta}_c$. Baseline *TVM* is correlated with five of the six characteristics: higher-baseline-*TVM* municipalities have higher pre-reform substance abuse, more RIPS service volume, lower poverty, lower public investment in health, and larger populations. Only the homicide rate is unrelated to baseline *TVM*. The joint test rejects balance at the 1% level ($F(6, 817) = 40.78, p < 0.001$). These correlations reflect the structural content of a shift-share exposure measure: municipalities with employment concentrated in tradable industries also tend to be larger, more urban, and to differ in healthcare-system capacity.

The level imbalance does not by itself violate the GPSS share-exogeneity assumption, which conditions on μ_m and λ_t . It does identify channels through which unobserved differential post-reform trends could enter the comparison. To address this, the specifications in Table 1 columns 2 and 4 and the default specifications in Tables 2–4 include controls selected by the [Belloni et al. \(2014\)](#) post-double-LASSO procedure.

Figure B2: Balance test of baseline municipality characteristics on the 2010s1 value of *TVM*



Note: Each marker reports the coefficient from a separate regression of the indicated standardized baseline municipality characteristic on the standardized 2010s1 value of *TVM*, with whiskers showing 95% confidence intervals from heteroskedasticity-robust standard errors. Regressions are weighted by 2009 working-age (18–65) population and run on the 824 municipalities in the substance-abuse estimation sample. Pre-reform substance-abuse rate and log RIPS service volume are measured in 2009s1; unmet basic needs, the share of public investment in health, and the homicide rate are measured in 2010; baseline population is the 2009 working-age (18–65) population. The dashed vertical line marks zero. The joint test of equal coefficients across all six characteristics is $F(6, 817) = 40.78, p < 0.001$.

C Data Appendix

Department-level Tariff Vulnerability Measure ($TVM_{d,t}$): The department-level analogue of $TVM_{m,y(t)}$ used in the Section 5.3 labor mechanism analysis. For department d and year-semester t , $TVM_{d,t}$ is a weighted average of two-digit SIC industry tariffs with weights given by the 2005 department-level tradable employment shares:

$$TVM_{d,t} = \sum_{j \in \mathcal{J}} \frac{\text{Emp}_{d,j,2005}}{\text{TotalEmp}_{d,2005}^{\text{tradable}}} \times \tau_{j,y(t)},$$

where $\text{Emp}_{d,j,2005}$ is 2005 employment in department d and industry j (aggregated across the municipalities in d), $\text{TotalEmp}_{d,2005}^{\text{tradable}}$ is the dept-level tradable employment total, and $\tau_{j,y(t)}$ is the same time-varying SIC tariff used in the $TVM_{m,y(t)}$ construction (annual values broadcast to both semesters of each year). The set $\mathcal{J}$ contains the 29 tradable 2-digit SIC rev. 3 industries (SIC 1, 2, 5, 10–36). The variable enters Table 5 and Figure 5.

Partner-tariff control ($z_{m,t}$): For municipality m and year t , the partner-tariff control is the employment-share-weighted mean of the tariffs that Colombia's main trading partners impose on Colombian exports:

$$z_{m,t} = \sum_{j \in \mathcal{J}} \frac{\text{Emp}_{m,j,2005}}{\text{TotalEmp}_{m,2005}^{\text{tradable}}} \times \bar{\tau}_{j,t}^{\text{part}},$$

where $\bar{\tau}_{j,t}^{\text{part}}$ is the unweighted mean across Colombia's main trading partners of the 2-digit SIC effectively applied tariff on industry j in year t , retrieved from WITS-WTO (partner-to-Colombia tariff schedules at the HS 6-digit level, aggregated to 2-digit SIC using the DANE crosswalk). The set $\mathcal{J}$ and the employment shares are those used in the TVM construction, so $z_{m,t}$ captures the time-varying market-access shock faced by the export sector employed in municipality m . The variable enters Appendix Table A4, column (2), in levels.

Input tariff: Following [Edmonds et al. \(2010\)](#) and [Chong and Velásquez \(2022\)](#), the municipality-year input tariff is built in two steps. First, for each tradable output

industry i , an industry-level input tariff is computed as the cost-share-weighted average of tariffs on industries supplying i :

$$IT_{i,t} = \sum_{j \in \mathcal{J}^{\text{in}}} sc_{j,i,2005} \times \tau_{j,t},$$

where $sc_{j,i,2005}$ is industry j 's share of industry i 's total intermediate input cost, taken from the 2005 Colombian input–output matrix published by DANE, and $\tau_{j,t}$ is the same SIC tariff used in the *TVM* construction. The input set $\mathcal{J}^{\text{in}}$ retains services on the input side (with $\tau_{j,t} = 0$, since services are not subject to tariffs) so that the cost shares are computed against the full intermediate-input bundle, following the construction of [Chong and Velásquez \(2022\)](#). The set of output industries i is restricted to the 23 tradable 2-digit SIC industries with a corresponding column in the Colombian input–output matrix. Second, the industry-level input tariff is aggregated to the municipality level using the same 2005 employment shares as *TVM*:

$$\text{InputTariff}_{m,t} = \sum_i \frac{\text{Emp}_{m,i,2005}}{\text{TotalEmp}_{m,2005}^{\text{tradable, IO}}} \times IT_{i,t}.$$

Here the sum runs over the 23 tradable industries with an input–output column, and $\text{TotalEmp}_{m,2005}^{\text{tradable, IO}}$ is total 2005 employment in those same 23 industries (not the full 29-industry tradable employment used in the *TVM* denominator), so the weights sum to one across i . The variable enters Appendix Table A4, column (3), in levels.

Baseline controls (candidate pool for the post-double-LASSO selection): Time-invariant baseline characteristics measured at the municipality level. Each variable enters the regression as a control $\times$ post-reform indicator. The four-variable candidate pool from which the [Belloni et al. \(2014\)](#) post-double-LASSO procedure selects the union for each outcome contains:

- Pre-reform substance-abuse patient rate per 100,000 working-age population (RIPS, 2009s1)
- Unsatisfied Basic Needs index (DANE NBI, 2005)

- Share of public investment in health (2010)
- Homicide rate per 100,000 population (Colombian National Police, 2010)

Log 2009 working-age (18–65) population (DANE projections) is added to this pool only in the augmented-pool sensitivity reported in Appendix Table A5.